Lower Limb:

Anterior Compartment of the Thigh

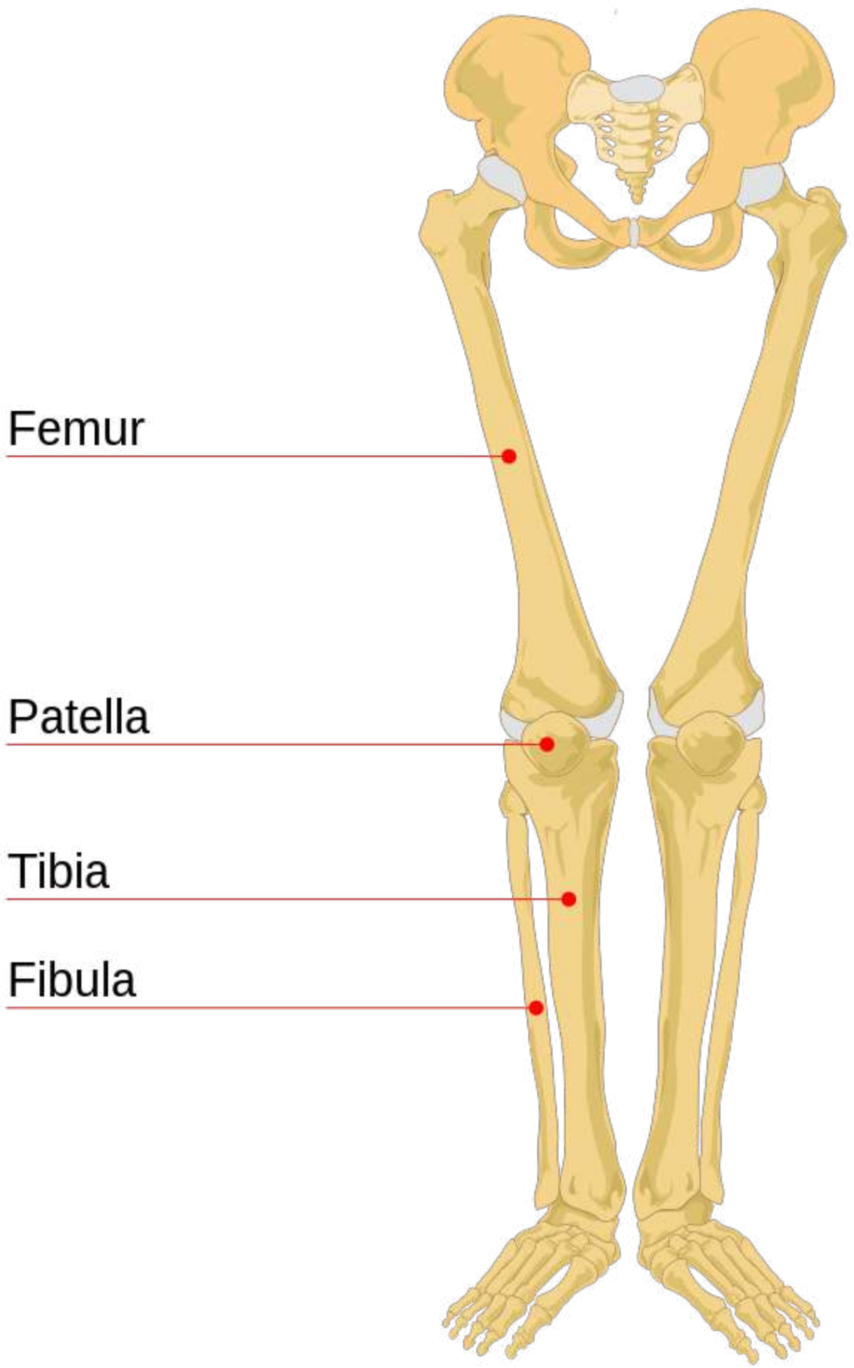
Dr.Nisreen Rajeh

Associate professor of anatomy

2025

First study the following bones of the lower limb before you study this lecture:

1. Pelvis
2. Femur
3. Patella
4. Upper tibia

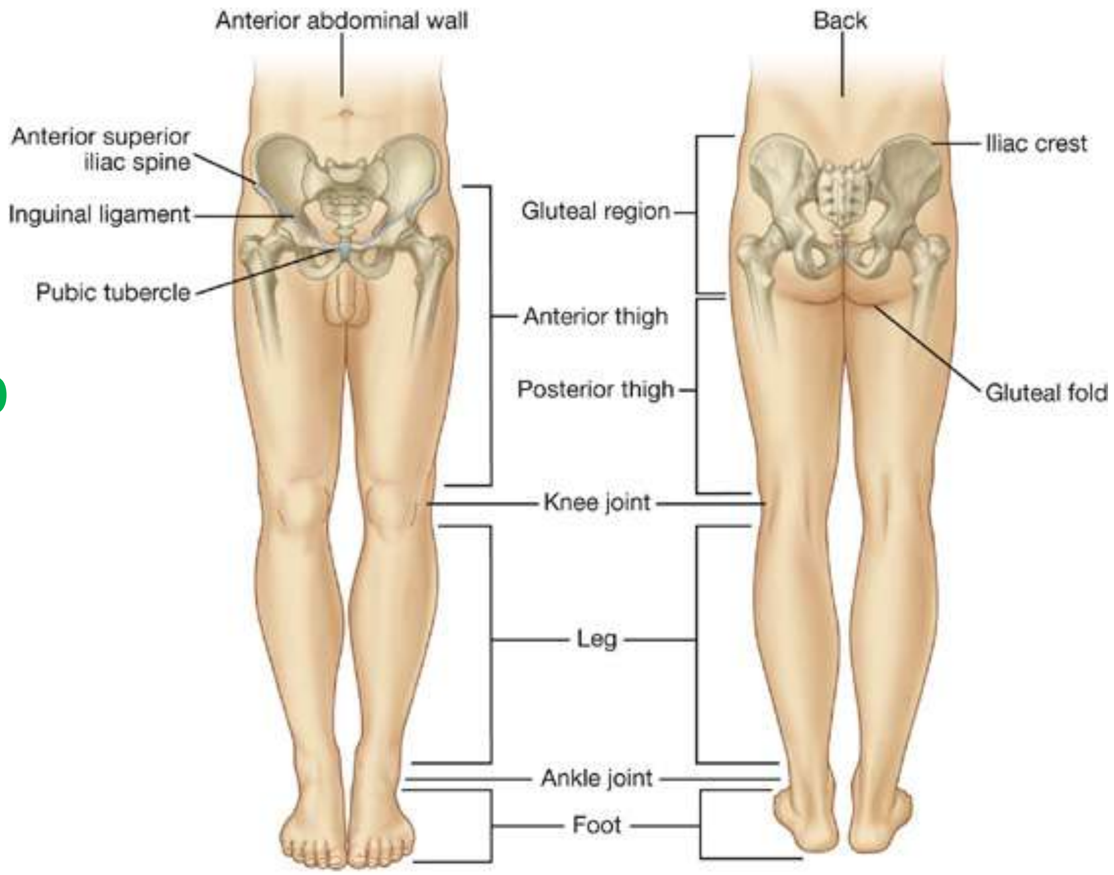


❖ Function of the lower limb:

1- Support body weight
2- Locomotion
3- Maintain balance

❖ Major six regions of the lower limb

- gluteal
- Femoral (thigh)
- Knee
- Leg
- Ankle
- Foot



Thigh: is the proximal segment of the lower limb, from the hip to the knee.

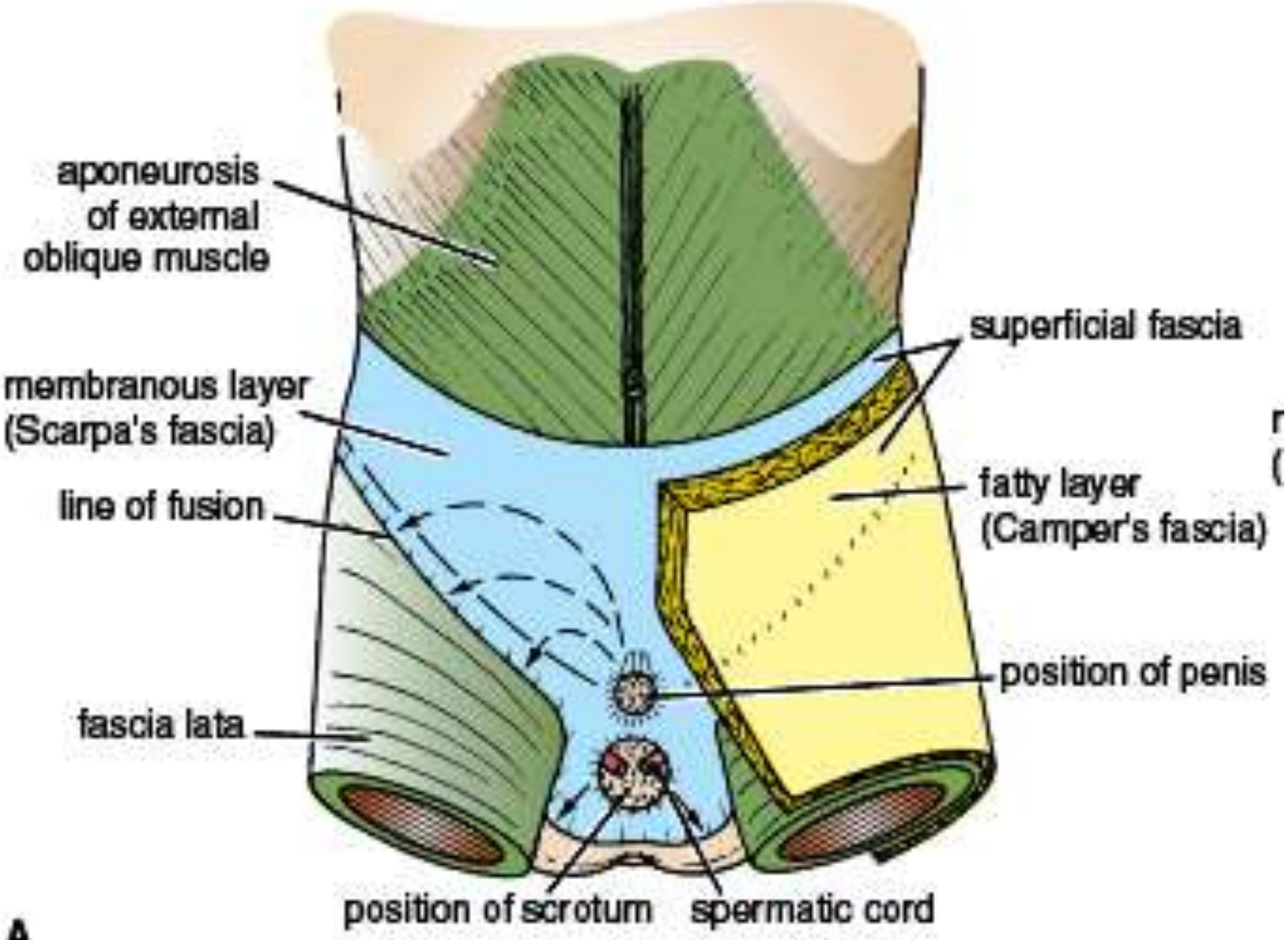
Skin & Fascia of the thigh

❖ **Fascia**: (difference between types of fascia?)(Nature of deep fascia of lower limb & why?)

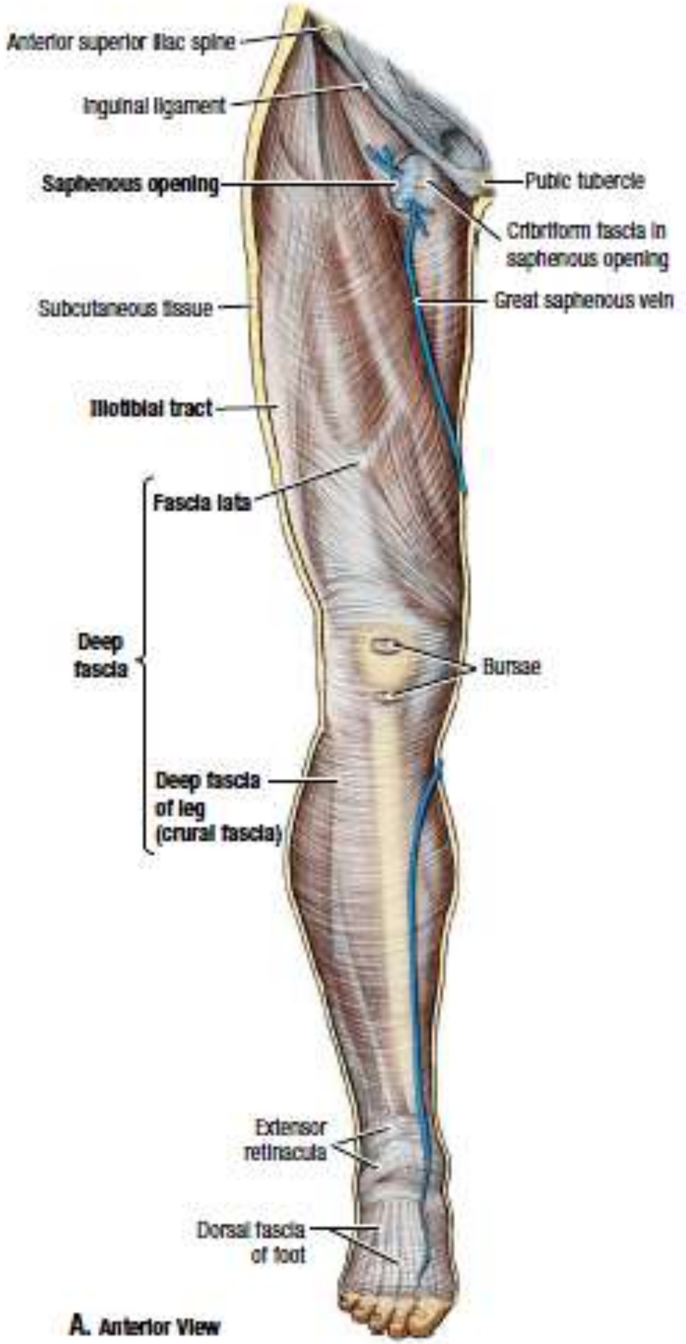
1- Superficial F.(fatty layer & membranous layer) loose CT, deep to skin, fat(extension of that on the ant.abdominal wall) , cutaneous nerves,superficial veins, lymph vessels & lymph nodes.

2- Deep F. (**Fascia lata**):

- **Nature?** Strong,llimits outward expansion of contracting muscles, increases venous returne to the heart with muscular contraction.
- **Attachment**;
  - Anteriorly: Inguinal ligament, pubic arch,body of pubis& pubic tubercle. And inferior abdominal wall membranous layer.
  - Posterolateral:. To iliac crest.
  - Posteromedial: To sacrum,coccyx and ischeal tuberosity.
  - Inferiorly : continous with   deep fascia of the leg below the knee.
  - **Laterally: thickened to form iliotibial tract.**
- **Hiatus? Saphenous opening**!

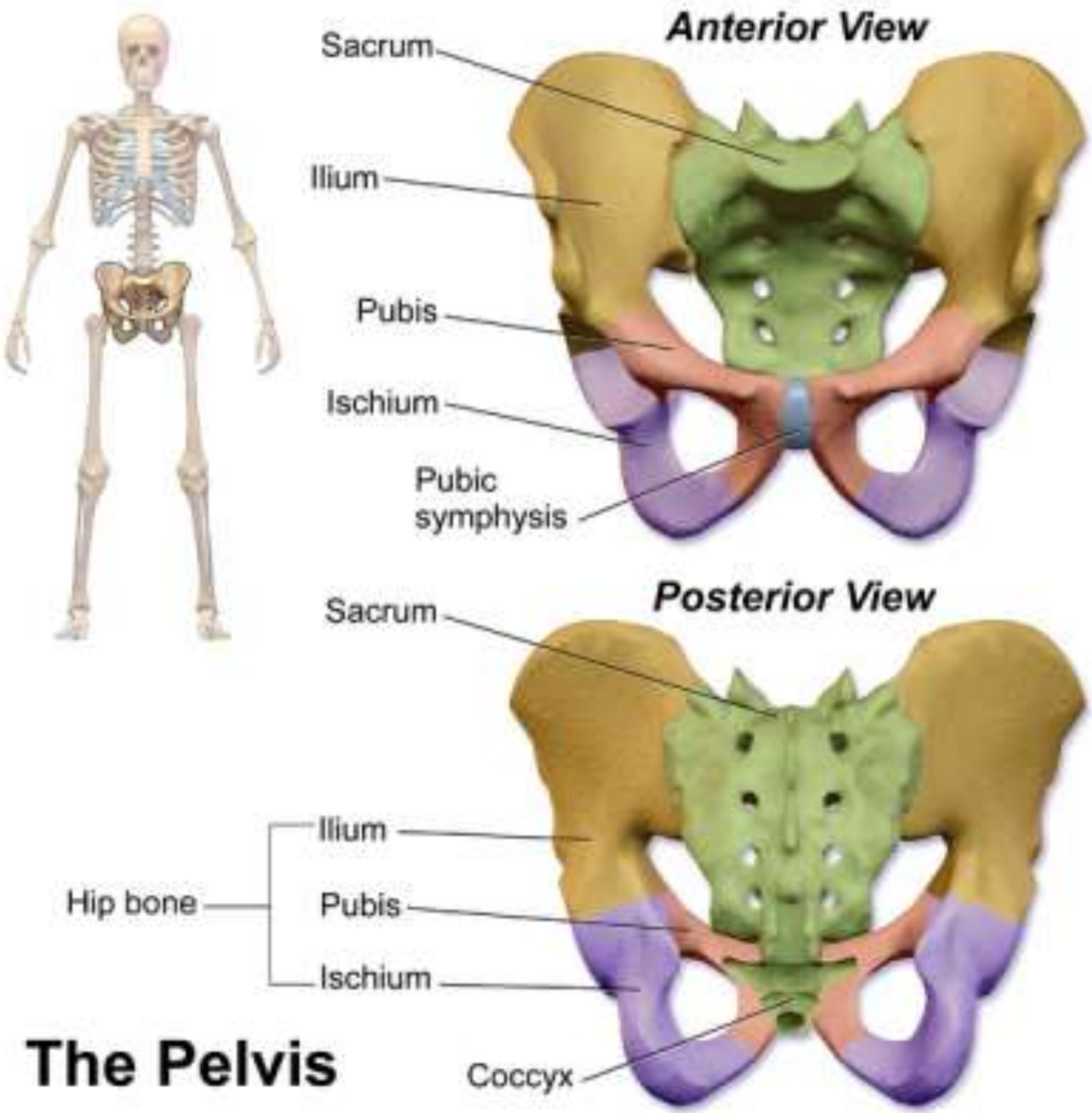

aponeurosis of external oblique muscle
superficial fascia
membranous layer (Scarpa's fascia)
line of fusion
fatty layer (Camper's fascia)
position of penis
fascia lata
position of scrotum
spermatic cord

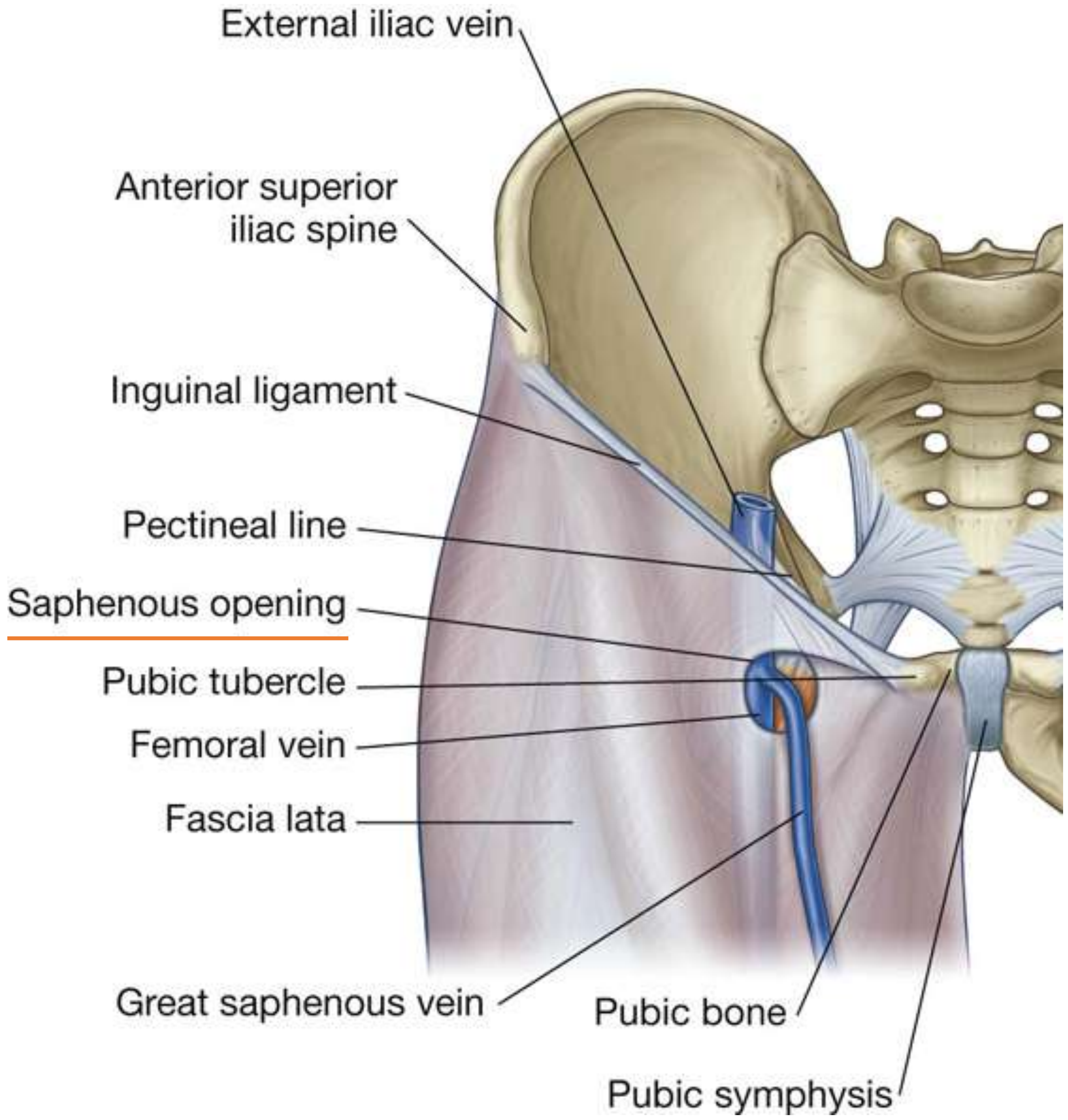
A



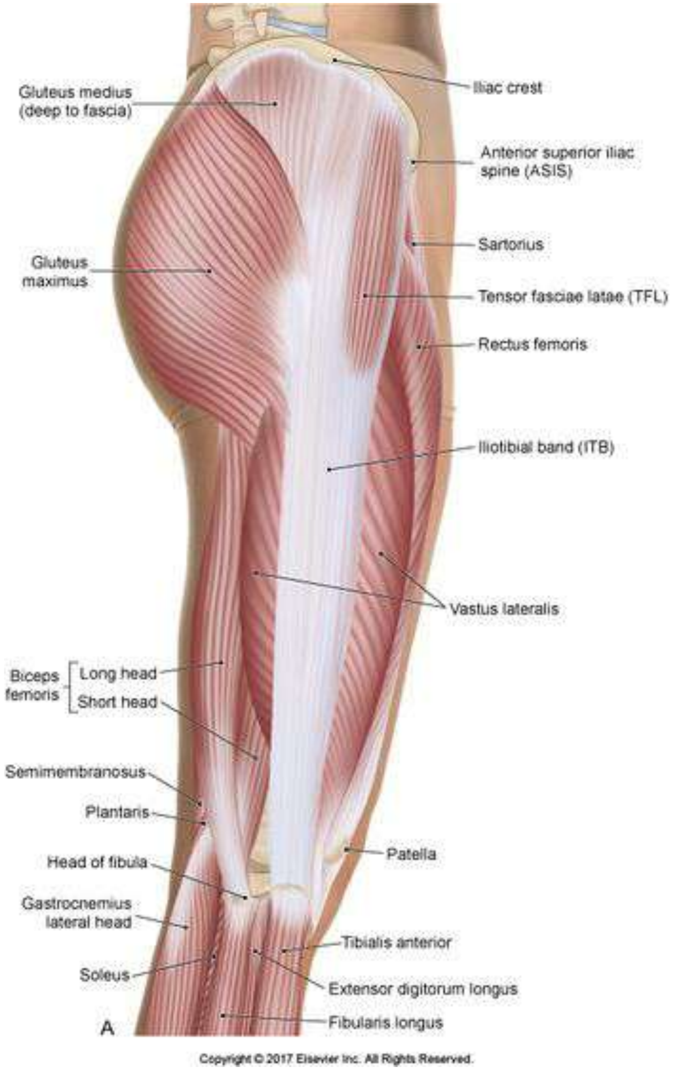
A. Anterior View

Fascia Lata


Anterior View
Sacrum
Ilium
Pubis
Ischium
Pubic
symphysis
Posterior View
Sacrum
Ilium
Hip bone
Pubis
Ischium
Coccyx
The Pelvis



- **Attachment of fascia lata to inguinal ligament.**



❑ Iliotibial tract:

- ➢ Definition?
- ➢ Site?
- ➢ Formation?
- ➢ Attachment?
- ➢ **Muscles inserted in?**
- ➢ Function?

Definition: Longitudinal fibrous sheath along lateral thigh, formed by thickening of fascia lata.

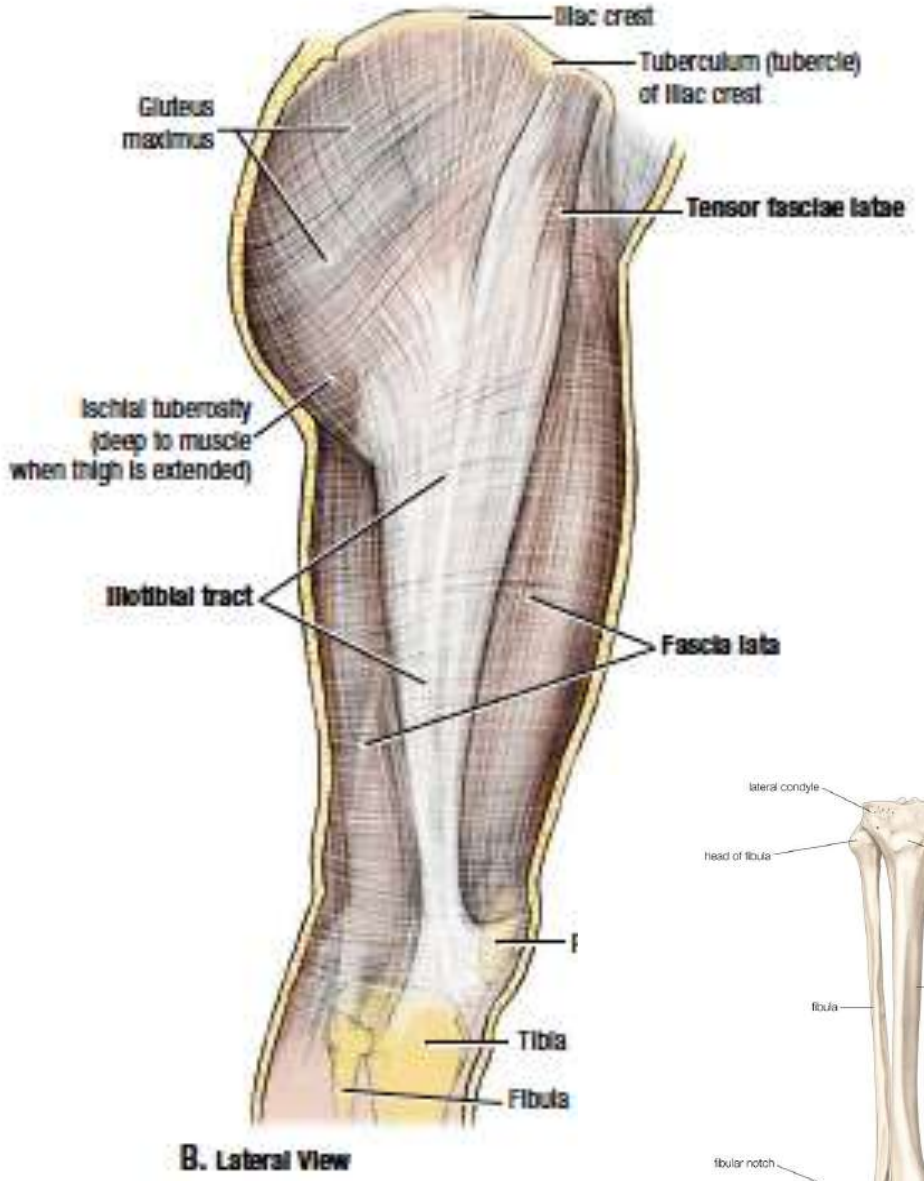
❖ Iliotibial tract

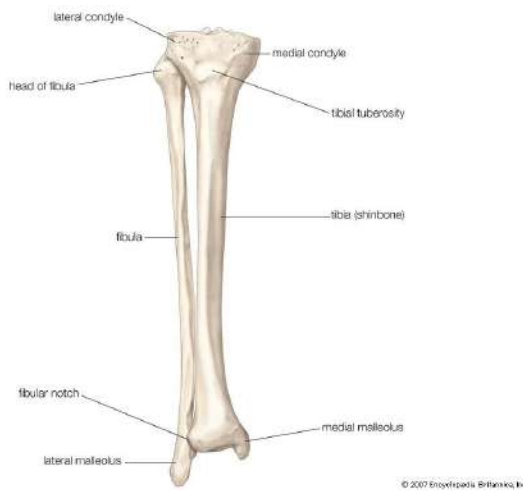
❖ **Attachment:**

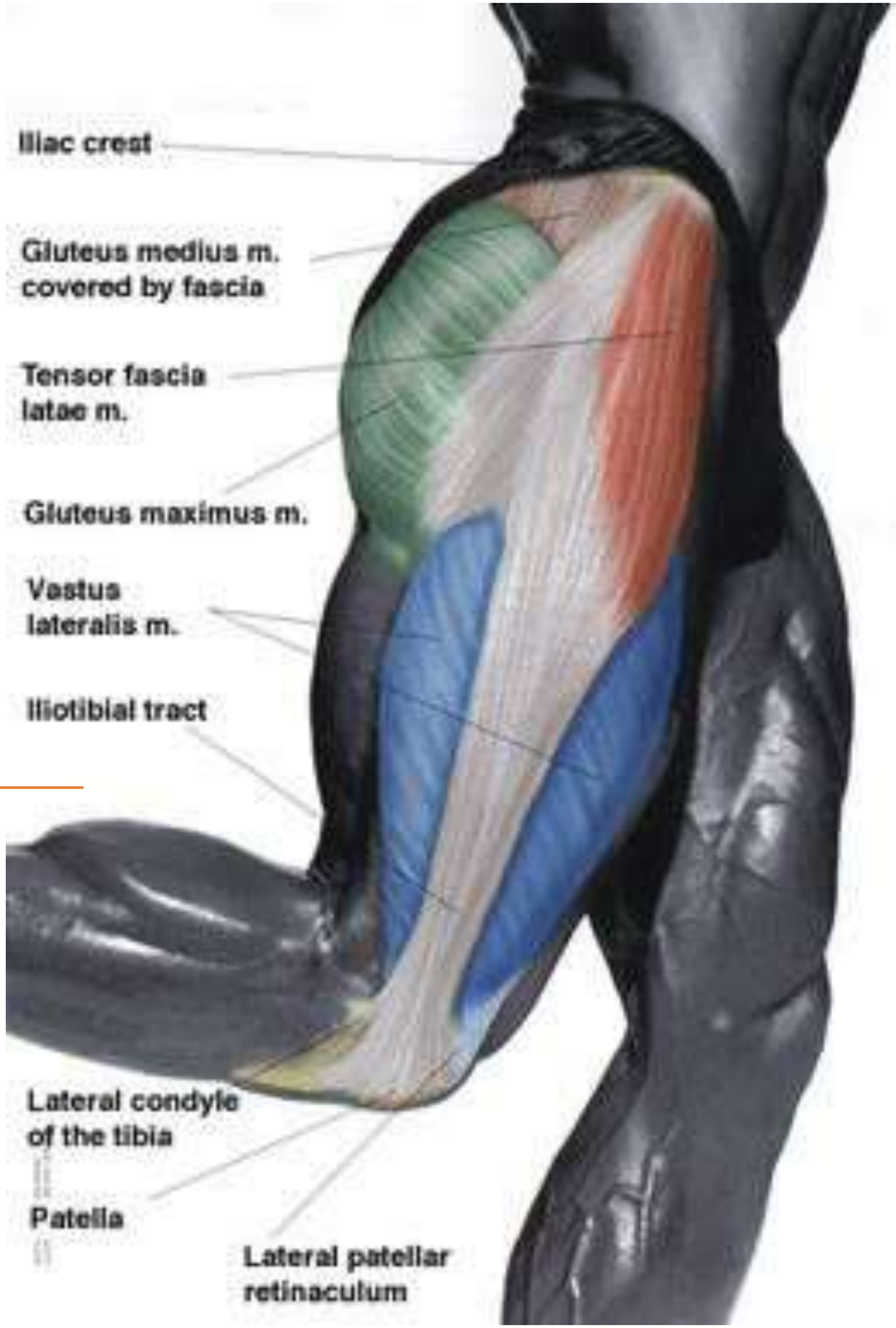
- **Above**: to iliac tubercle.

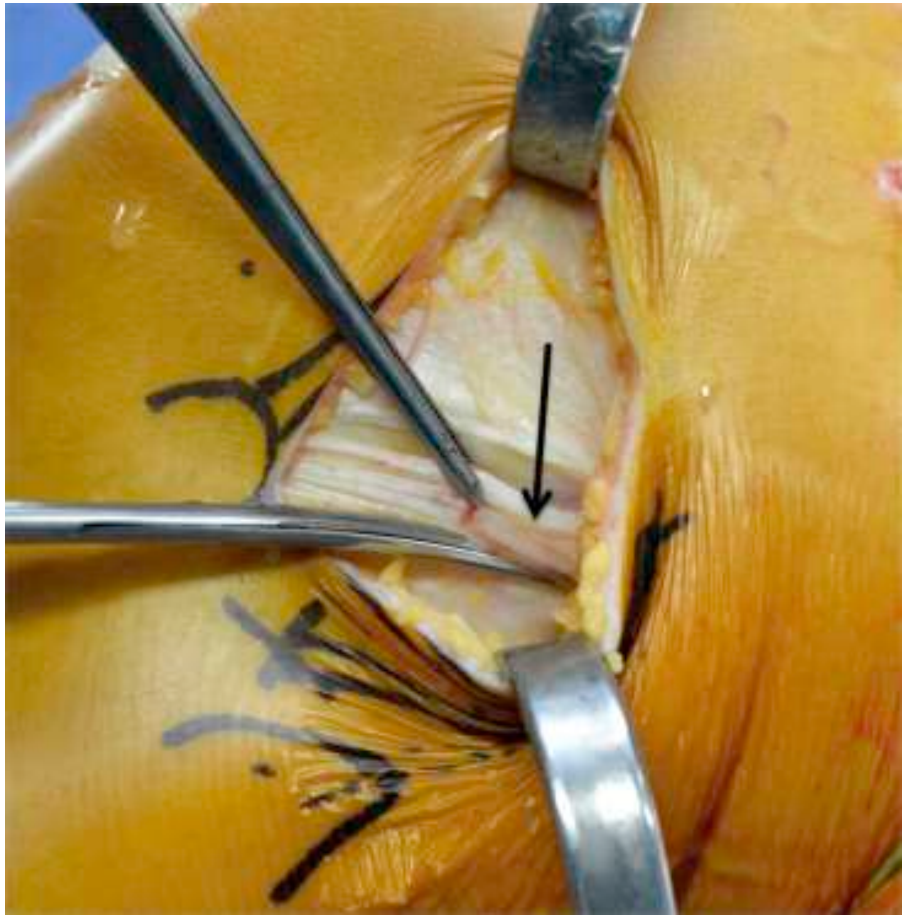
- **Below**: lateral condyle of tibia.

مهم **It receives insertion of tensor fasciae latae & greater part of gluteus maximus muscle**.






Iliac crest
Gluteus medius m. covered by fascia
Tensor fascia latae m.
Gluteus maximus m.
Vastus lateralis m.
Iliotibial tract
Lateral condyle of the tibia
Patella
Lateral patellar retinaculum

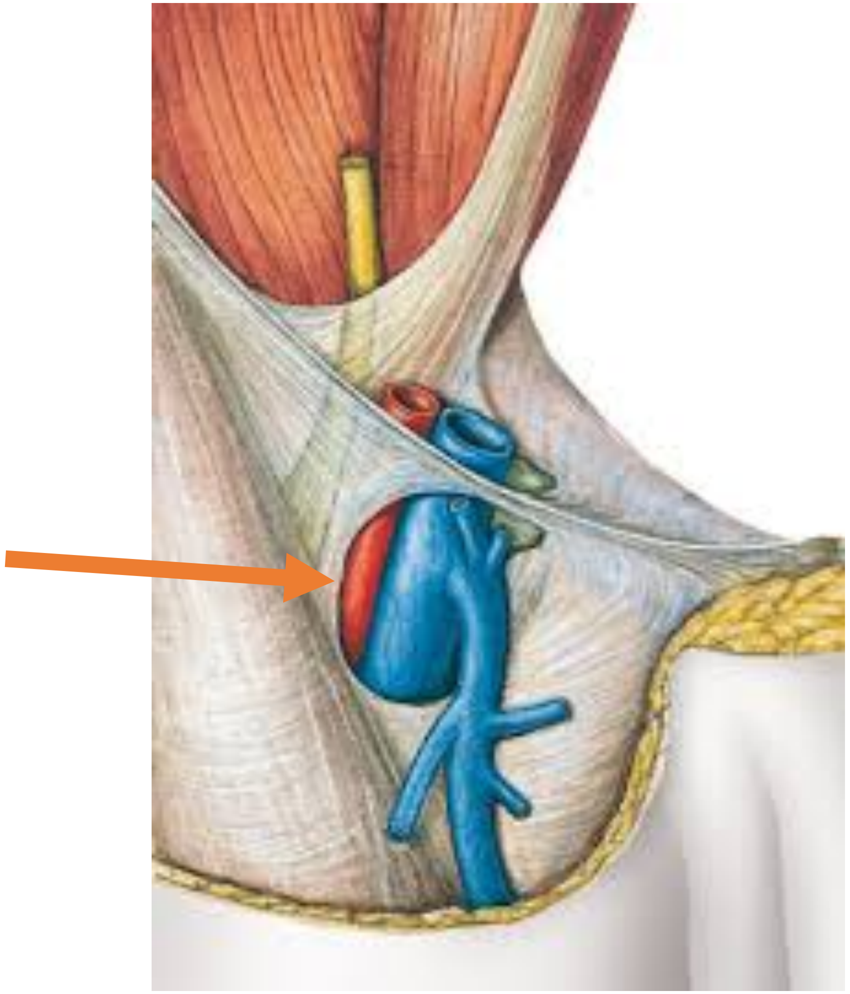


❑ Functions of Iliotibial tract or band:

- ITB plays a role in the transmission of forces from these two muscles across both the **knee and the hip**.

- Its main functions are **pelvic stabilization and posture control & assists in knee extension**.

(Saphenous Opening)

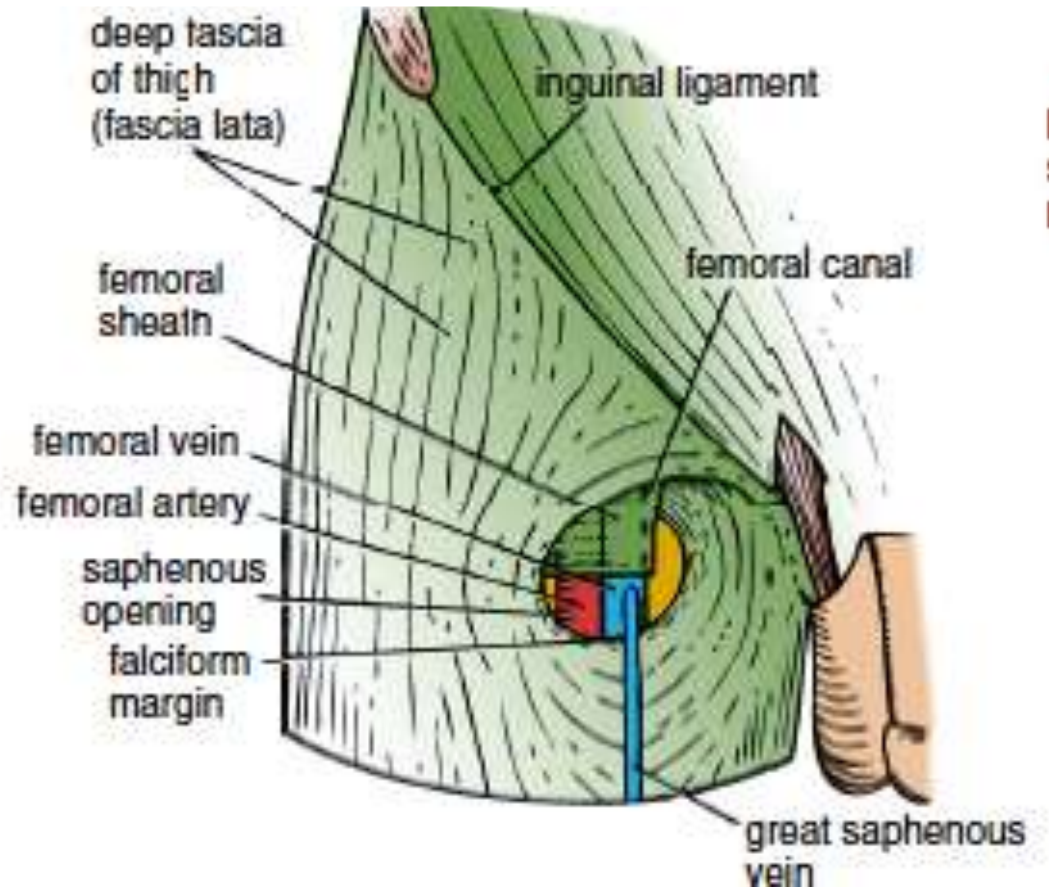


- Definition?
- Site?
- Content &importance?

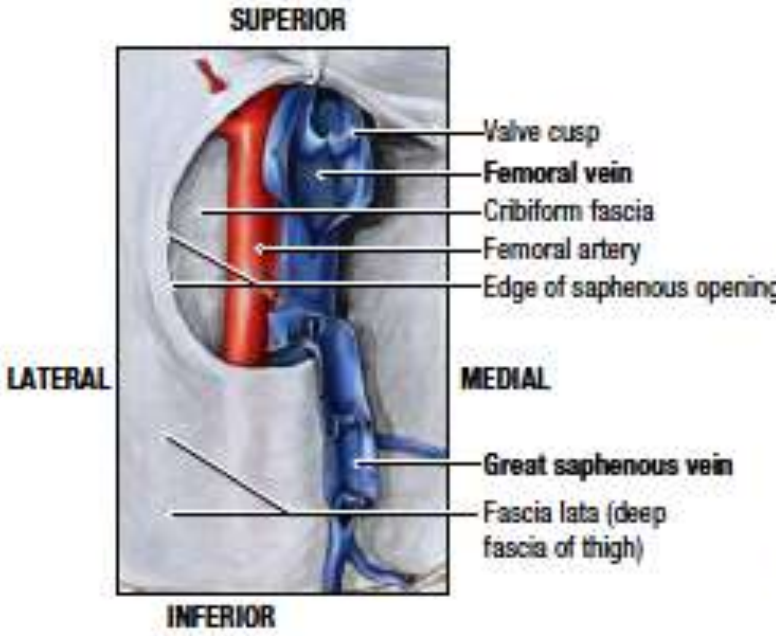
❖ <u>Saphenous opening</u>

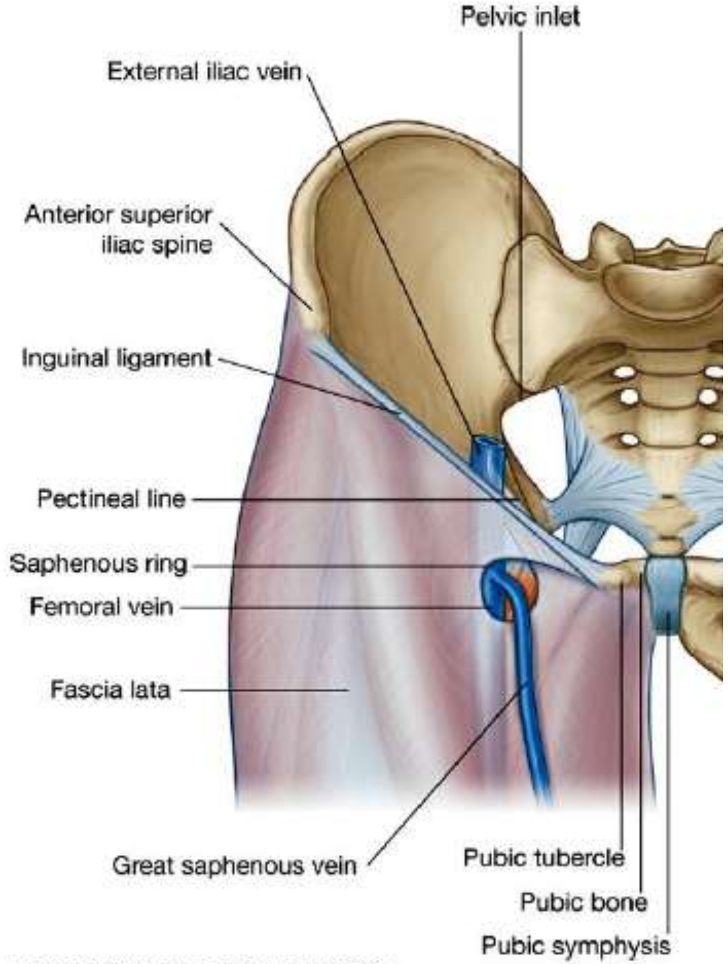
❖Is an oval opening in the **deep fascia** in front of the thigh just below the inguinal ligament.

❖<u>Site: </u>1.5 inch below& lateral to **pubic tubercle**.



- <u>**Importance:**</u> it transmits the **great saphenous vein (to end in femoral vein)**, some small **branches of femoral artery** and **lymph vessels.**

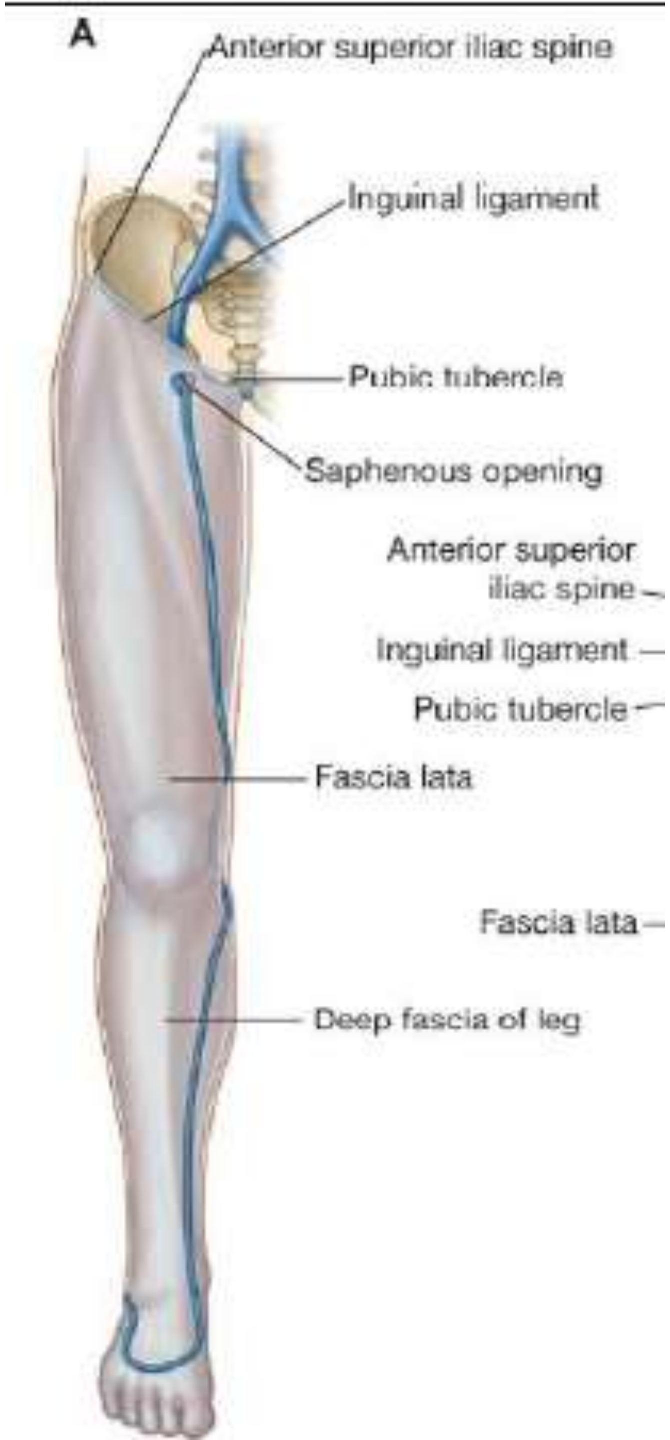




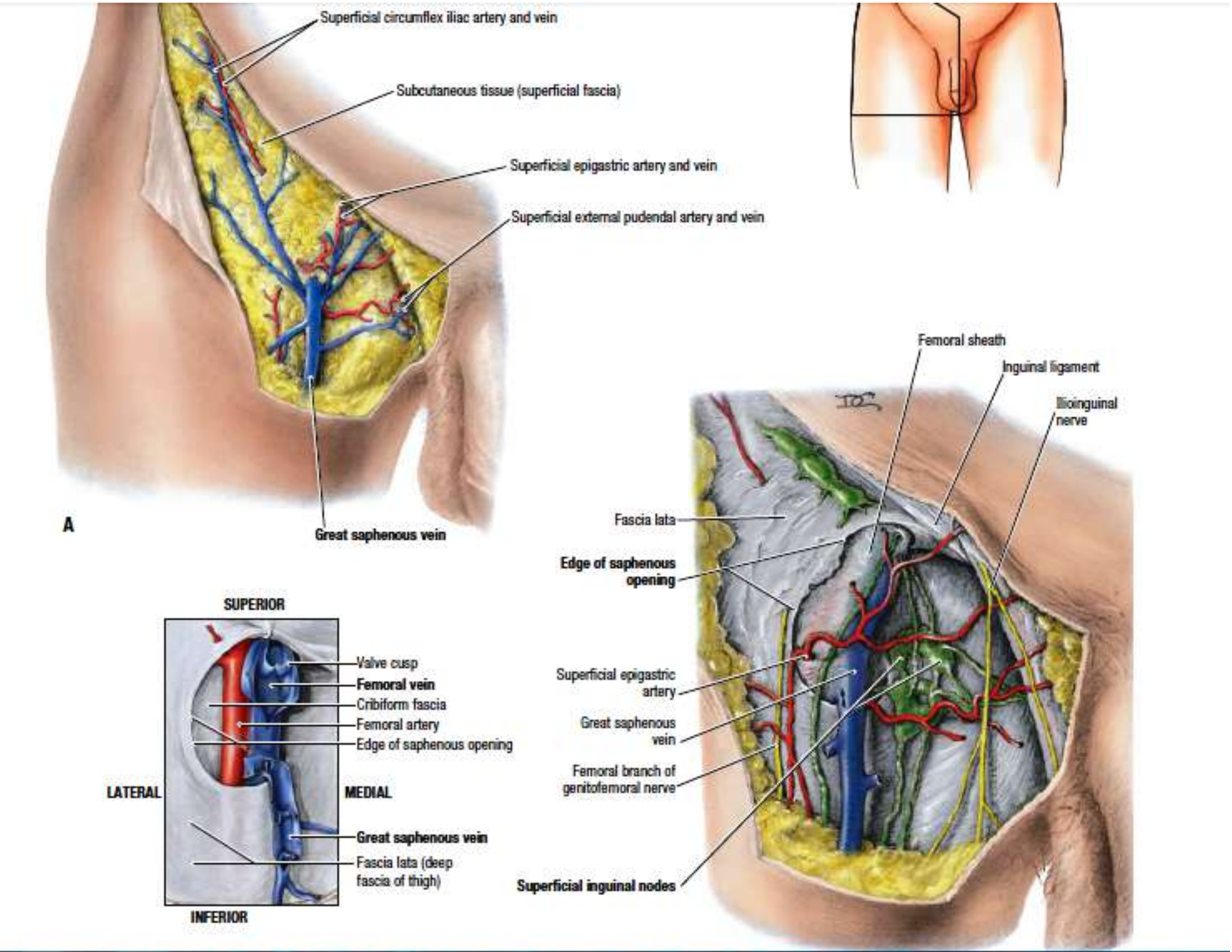
Drake: Gray's Anatomy for Students, 2nd Edition.

Copyright © 2009 by Churchill Livingstone, an imprint of Elsevier, Inc. All rights reserved

Figure 6.40 Saphenous ring. Anterior view.



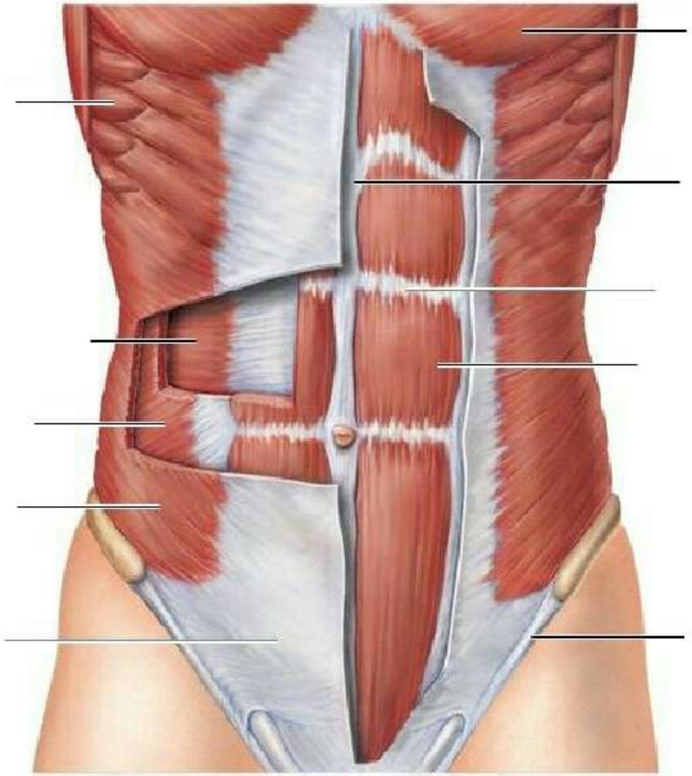
C. Anteromedial View, Normal Veins



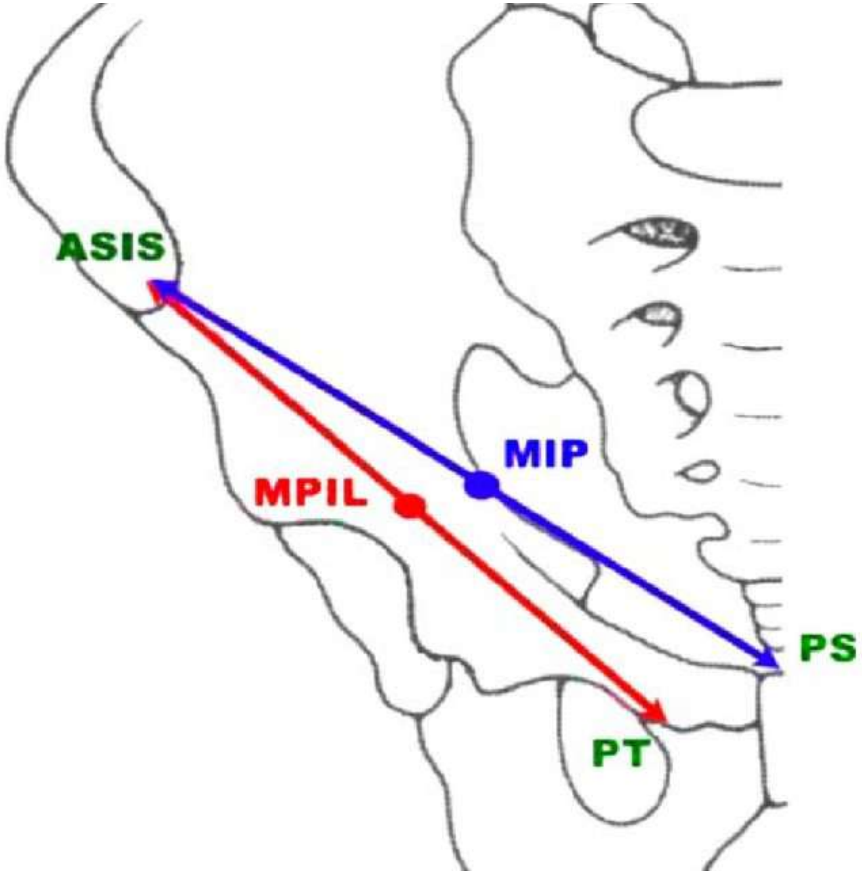
Superficial circumflex iliac artery and vein
Subcutaneous tissue (superficial fascia)
Superficial epigastric artery and vein
Superficial external pudendal artery and vein
A
Great saphenous vein
SUPERIOR
Valve cusp
Femoral vein
Cribiform fascia
Femoral artery
Edge of saphenous opening
LATERAL
MEDIAL
Great saphenous vein
Fascia lata (deep fascia of thigh)
INFERIOR
Femoral sheath
Inguinal ligament
Ilioinguinal nerve
Fascia lata
Edge of saphenous opening
Superficial epigastric artery
Great saphenous vein
Femoral branch of genitofemoral nerve
Superficial inguinal nodes

(Inguinal Ligament)

- Important landmark & gate ;**anatomically** **Why?**
- **Definition:** it is the lower border of aponeurosis of **external oblique muscle** of the abdomen, folded backwards upon itself, forming the boundary between the anterior abdominal wall & the front of thigh.
- **Attachment:** laterally: to ant.sup.iliac spine (ASIS).
  Medially: pubic tuercle.
- Structures pass deep to it: iliacus,psoas, femoral nerve, femoral sheath, femoral vessels, lymph vessels & lat. Cut. Nerve of the thigh.
- **Midpoint of inguinal ligament (MPIL):** is the point on the inguinal ligamnet midway between the ant.sup.iliac spine and the pubic tubercle= **middle of the inguinal ligamnet.**
- **Midinguinal point (MIP):** is the point on the inguinal ligamnet midway between the ant.sup.iliac spine and the symphesis pubis.



EXTERNAL OBLIQUE ABDOMINAL MUSCLE



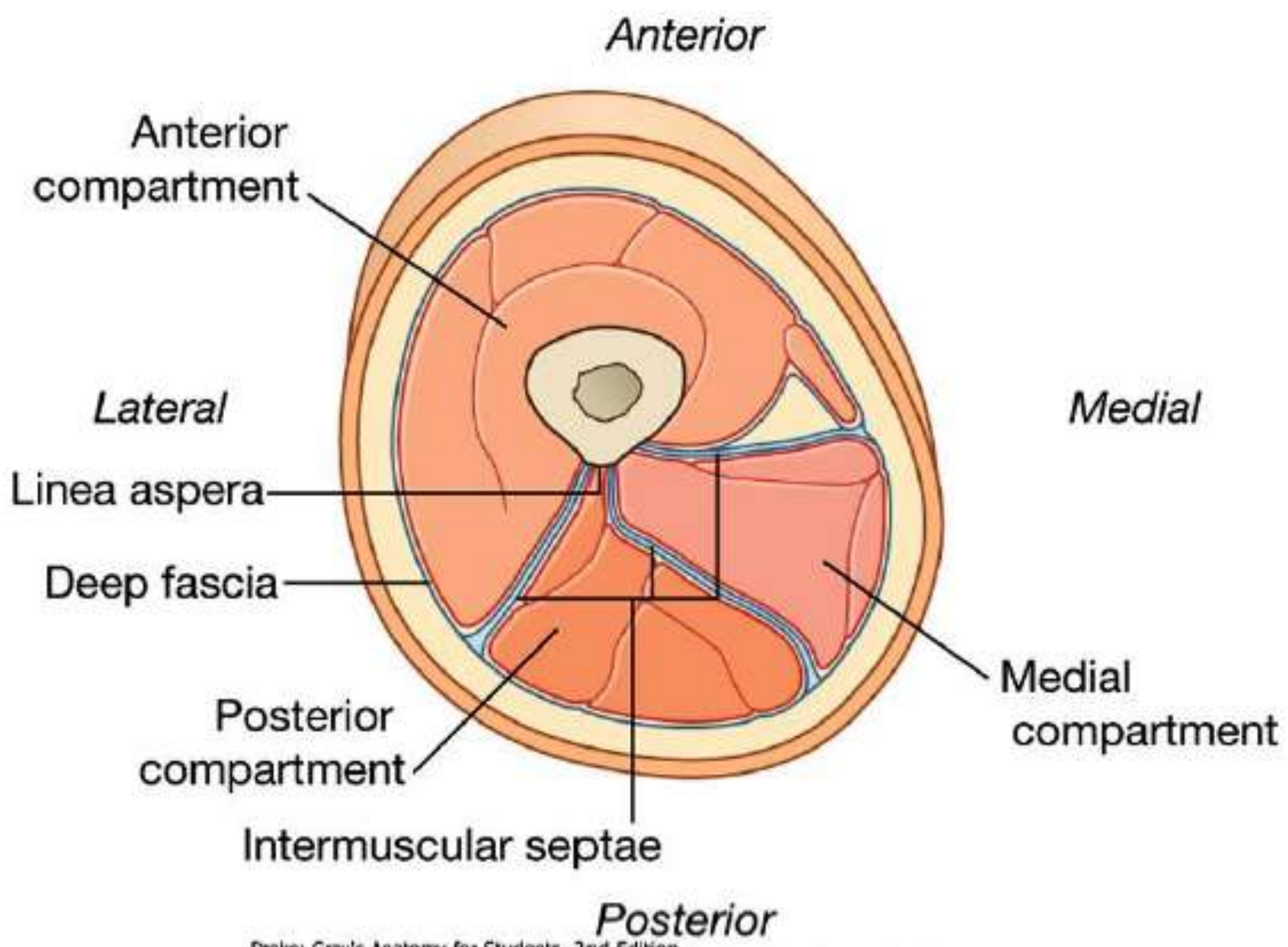
❑ Fascial compartments of the thigh:

- ❑ Three fascial septa pass from deep fascia to linea aspera of the femur.
- ❑ So the thigh will be divided into three compartments (muscles + nerves+ vessels).
- ❑ Compartments are:

1-Anterior (flexors for the hip & extensors for the knee)

2-Medial (adductors of thigh)

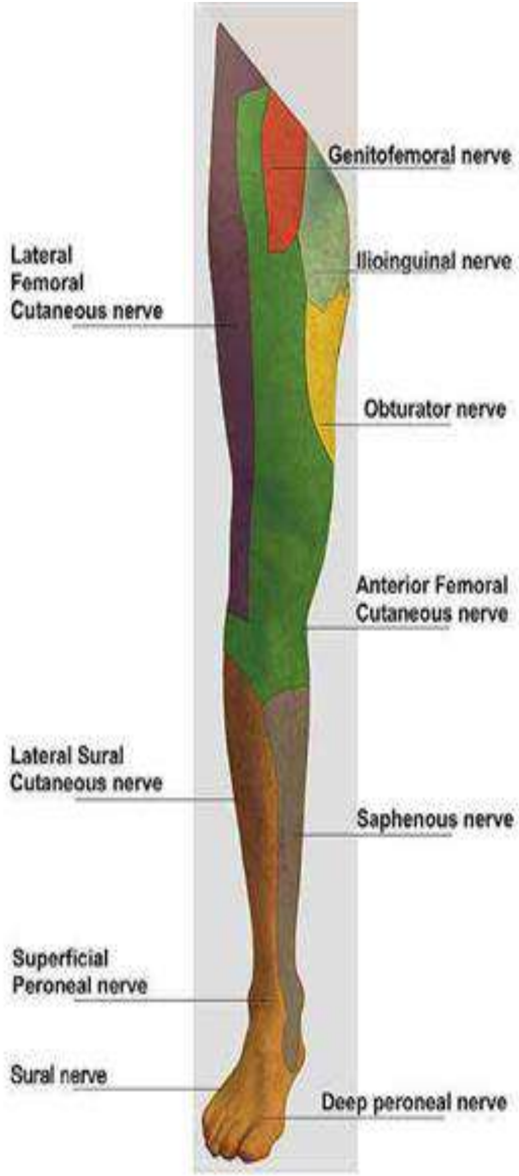
3-Posterior (extensors for the hip and flexors for the knee)



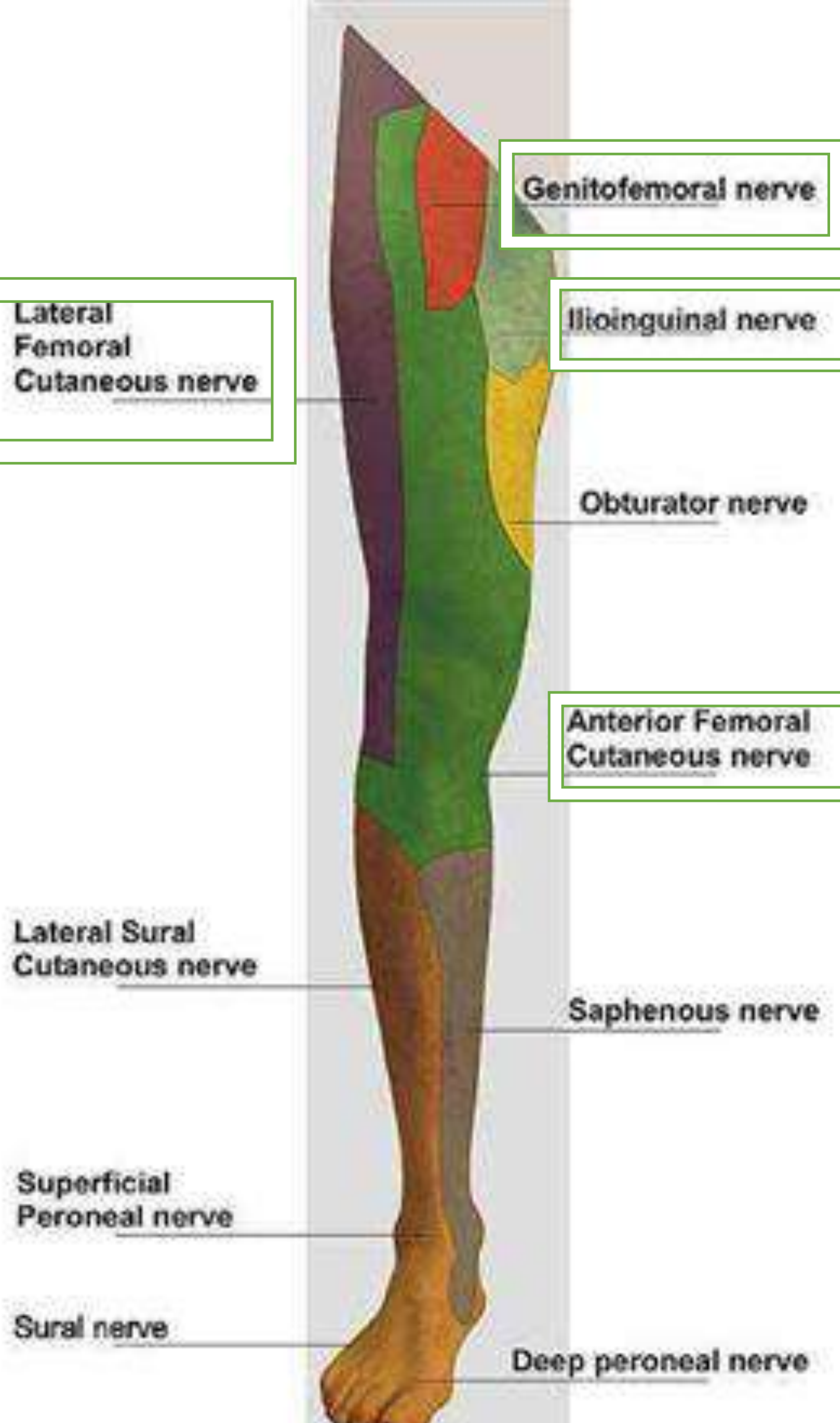
Drake: Gray's Anatomy for Students, 2nd Edition.
Copyright © 2009 by Churchill Livingstone, an imprint of Elsevier, Inc. All rights reserved.
Figure 6.55 Transverse section through the midthigh.

Cutaneous nerves of the anterior and lateral thigh:

Cutaneous nerve	Origin	Area of skin supply in thigh
1. Ilioinguinal nerve	Branch of **lumbar plexus (L1).**	Skin of **small area below the Medial part of inguinal ligament.**
2.Genitofemoral nerve(femoral branch) .	Branch of **lumbar plexus (L1,L2)**	Enters the thigh behind the **middle of inguinal ligament,**supplies **most of the skin over femoral triangle.**
3-Intermediate cutaneous nerve of the thigh.	Branch of **femoral nerve**	To skin of **Anterior aspect of the thigh till the knee.**

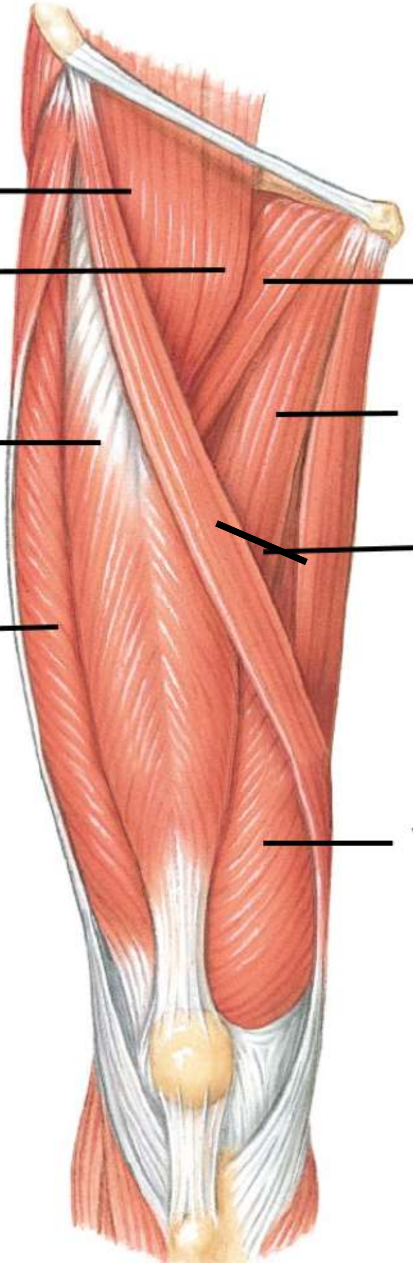


Cutaneous nerve	Origin	Area of skin supply in thigh
4. Patellar plexus.	➢ Lies in:superficial fascia covering front of patella. ➢ formed of terminal branches of : ➢ **Medial & intermediate** cutaneous nerves of the thigh. ➢ **Lateral cutaneous** nerve of the thigh. ➢ Infrapatellar branch of **saphenous nerve**.	To skin covering **patella & ligamentum patellae.**
5-Lateral cutaneous nerve of the thigh.	Branch of **lumbar plexus (L2,L3)**	Skin of the **lateral aspect of the thigh & knee.**


Genitofemoral nerve
Lateral Femoral Cutaneous nerve
Ilioinguinal nerve
Obturator nerve
Anterior Femoral Cutaneous nerve
Lateral Sural Cutaneous nerve
Saphenous nerve
Superficial Peroneal nerve
Sural nerve
Deep peroneal nerve

Anterior fascial compartment contents

- **Muscles (7):** 1-Sartorius, 2-Iliopsoas,
  3-pectineus, 4- quadriceps femoris;
  (Rectus femoris+ Vastus lateralis+ Vastus medialis+ Vastus intermedius)
- **Blood supply:** Femoral artery.
- **Nerve supply:** Femoral nerve.



2
Iliacus
3
Psoas major
Pectineus
4
Adductor longus
Rectus femoris
Sartorius
1
Vastus lateralis
Vastus medialis

1.Sartorius (tailor"s muscle)(narrow, strap-shaped muscle)

O: from anterior ***superior*** iliac spine, run downward & medially.

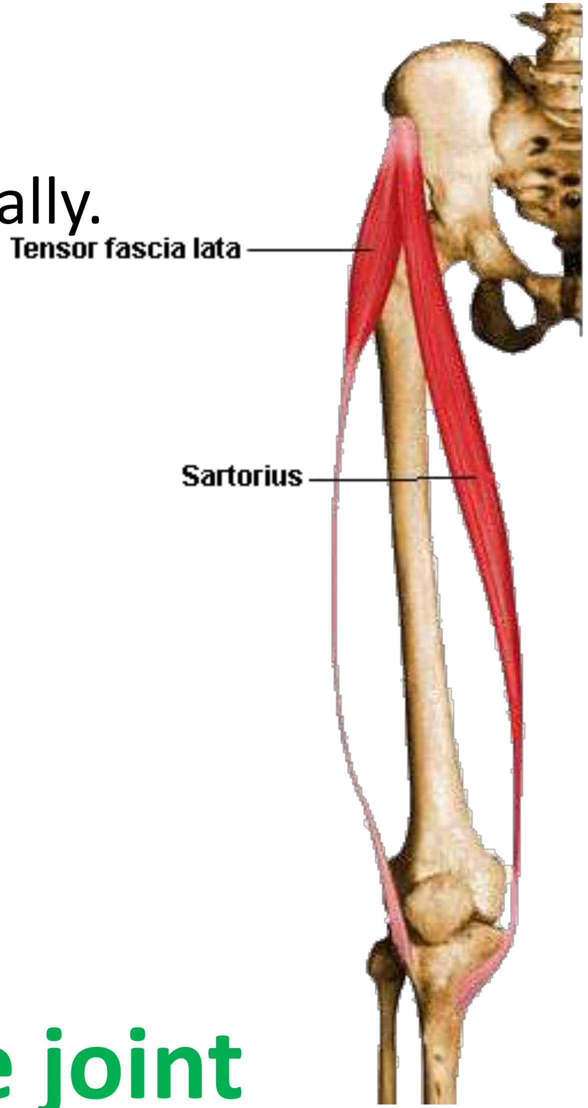
I: upper part of medial surface of shaft of tibia**(S.G.S.).?**

NS: **Femoral nerve**

A: acts on two joints:

1- hip joint: flexion, abduction& lateral rotation of thigh at hip joint.

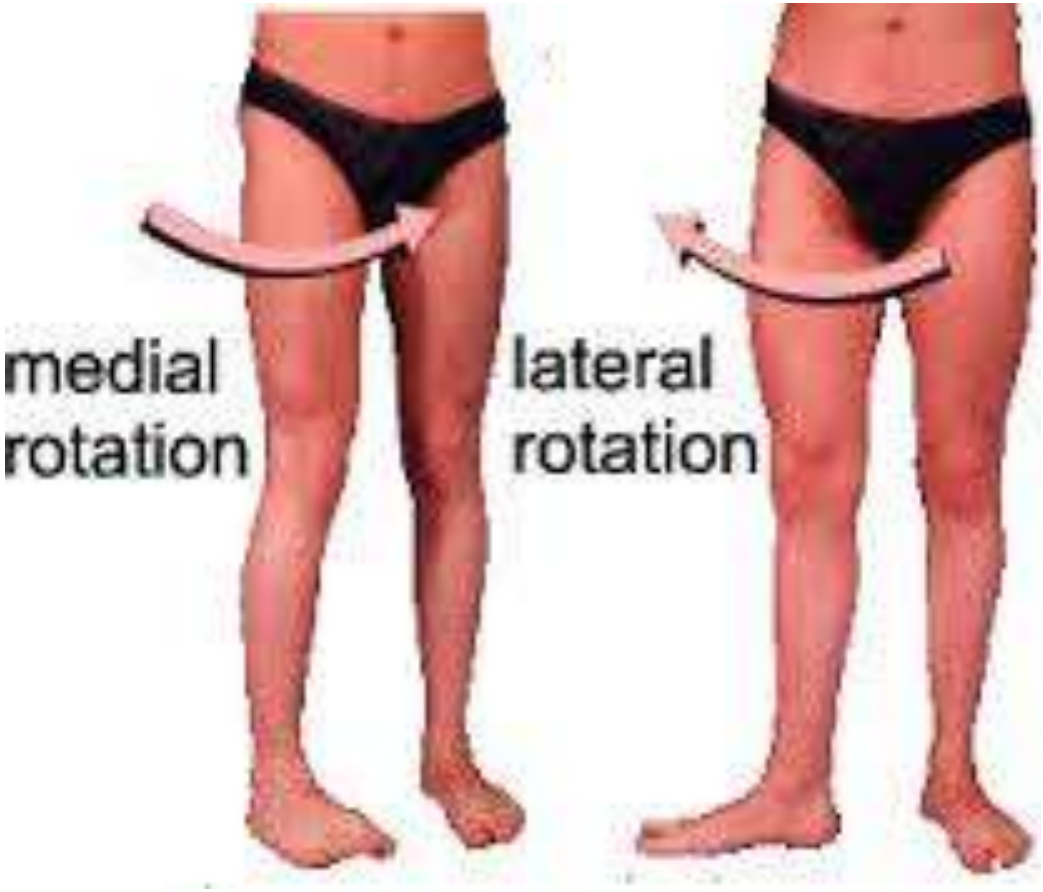
2- Knee joint: flexion & medial rotation of Leg at knee joint.

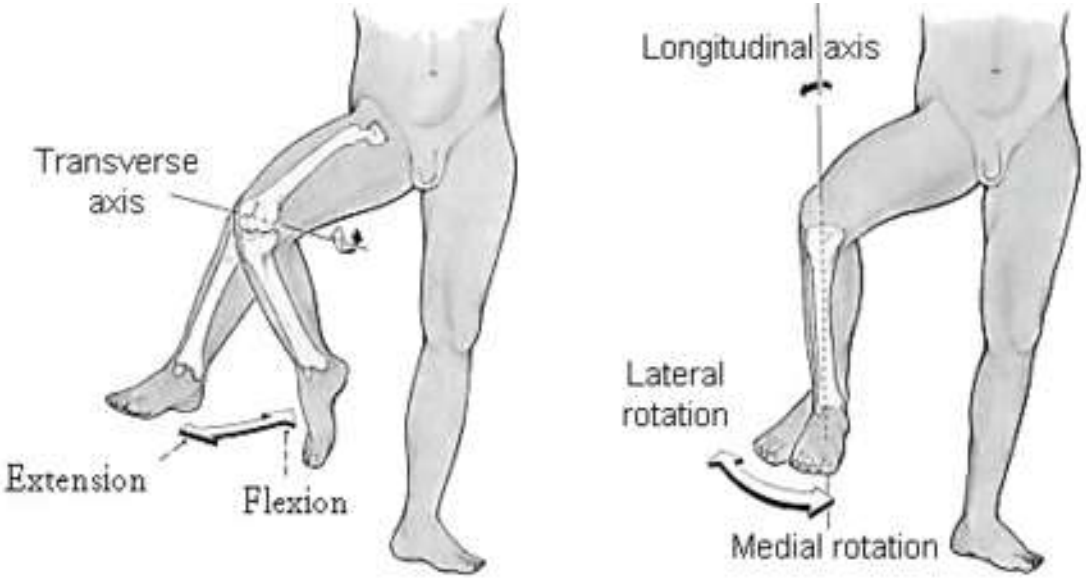


N.B: it is a flexor for both hip and knee joint

- The action of **both sartorius** muscles; cross leged sitting position.


medial
rotation
lateral
rotation


Transverse
axis
Extension
Flexion
Longitudinal axis
Lateral
rotation
Medial rotation


137


Sartorius

2)Iliacus (fan-shaped muscle)

O: Iliac fossa of hip bone within the abdomen.

I: the fibers converge& joint the tendon of the psoas to form iliopsoas muscle.iliopsoas tendon inserted in **lesser trochanter** of femure.

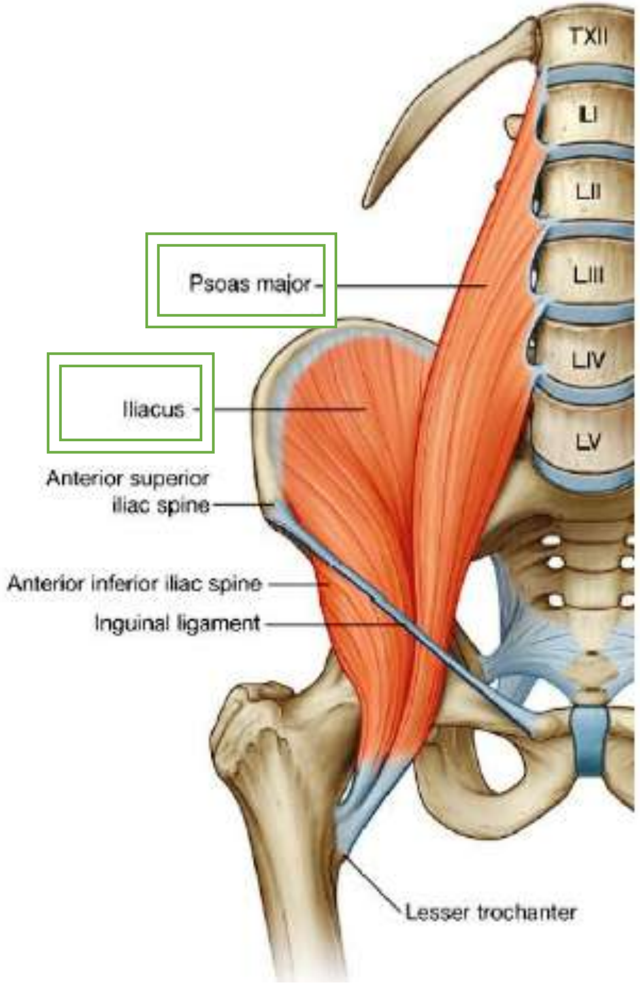
Ns: A **branch of femoral nerve** within the abdomen.

A: iliopsoas action:

1)flexes the thigh on the trunk at the hip joint.

2) if the thigh is fixed, it flexes the trunk on the thigh.

3)It also **medially rotates** the thigh.



3)Psoas major

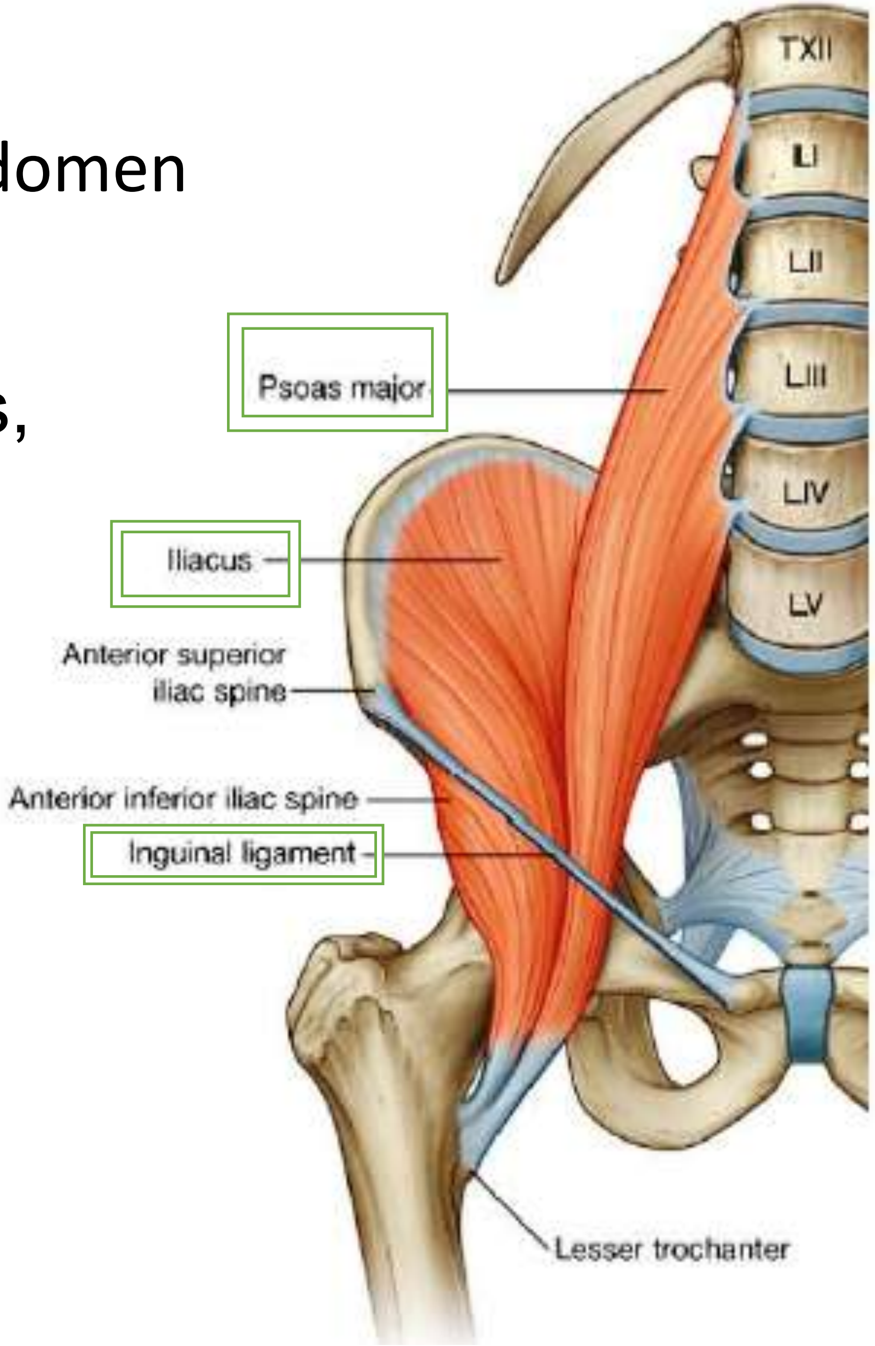
- Long, fusiform muscle arises within the abdomen and descends into the thigh.

O: **T12- L5** vertebrae (transverse processes, sides of vertebral bodies& intervertebral discs).

I: its fibers leave the abdomen by passing behind inguinal ligament. iliopsoas tendon inserted in **lesser trochanter** of femure.

NS: branches from Lumbar plexus.

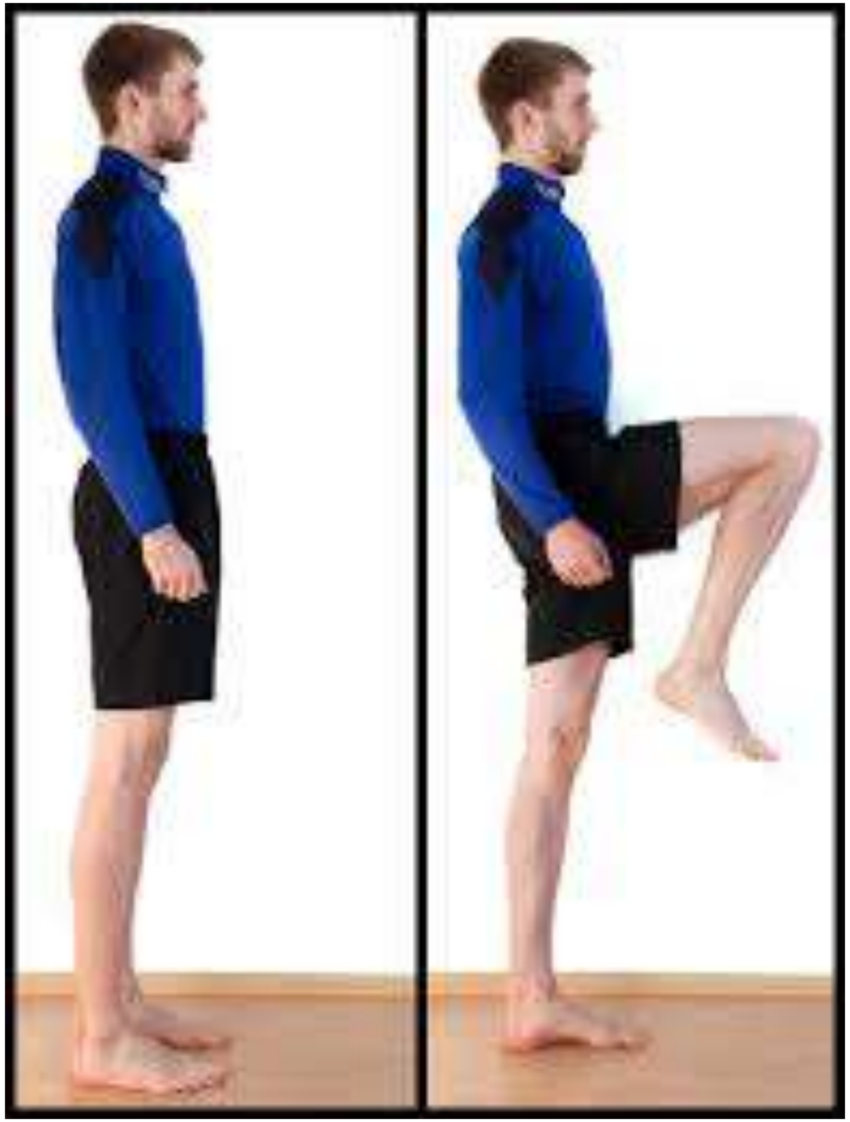
A: see iliopsoas action.

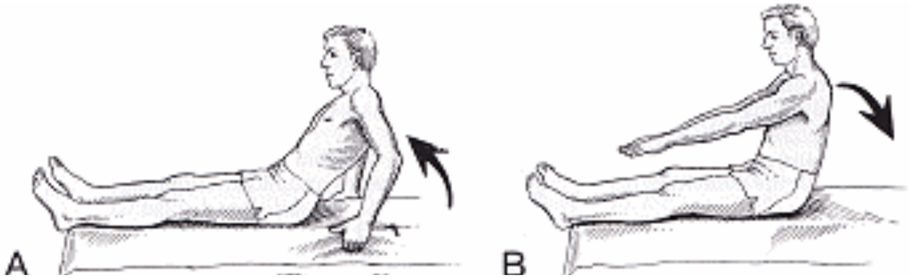


- Posas major:The chief flexor of the thigh, the most powerful of the hip flexors with the longest range .

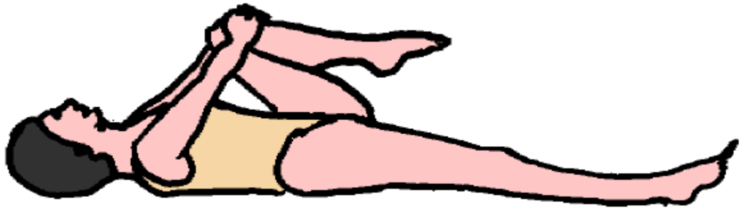
- It is the only muscle attached to vertebral column, pelvis and femur.

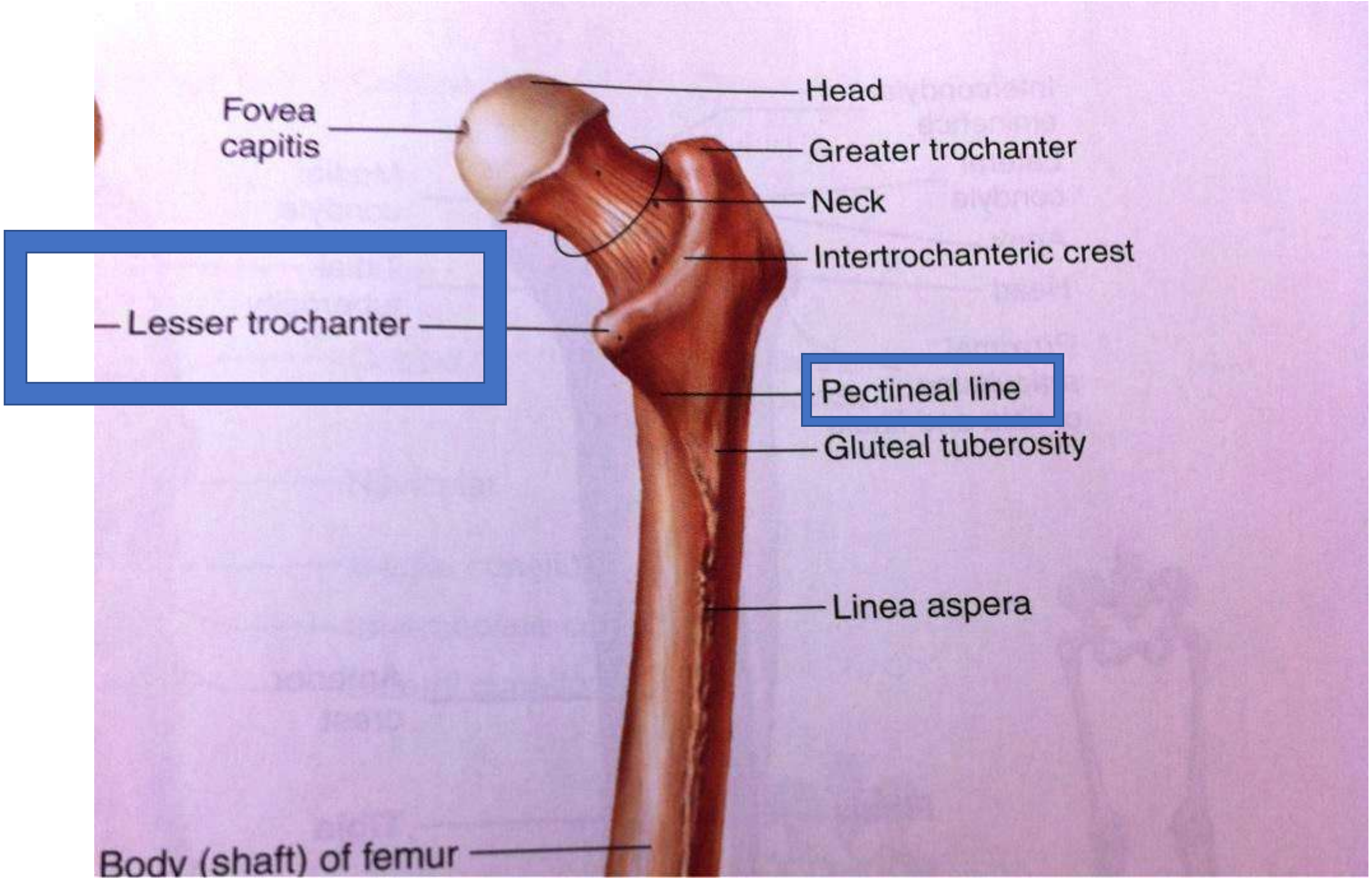
Iliopsoas action:





Flexes thigh on trunk; if thigh is fixed,it flexes the trunk on thigh as in sitting up from lying position.

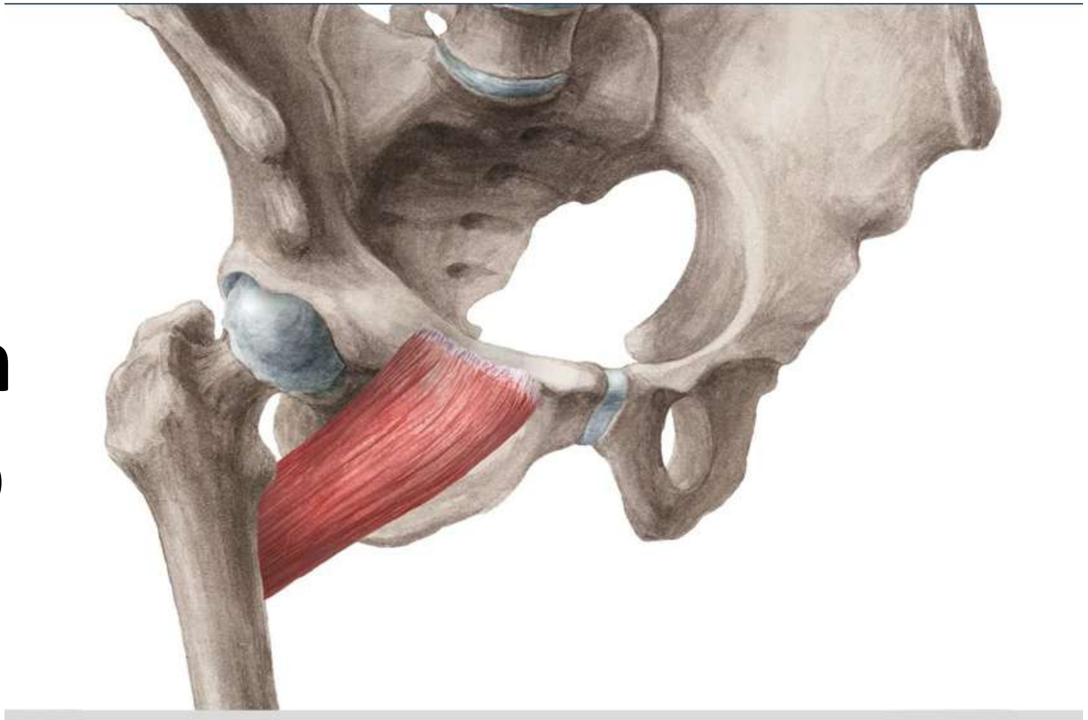



Head
Fovea capitis
Greater trochanter
Neck
Intertrochanteric crest
Lesser trochanter
Pectineal line
Gluteal tuberosity
Linea aspera
Body (shaft) of femur

4) Pectineus; (quadrangular muscle)

O: Superior ramus of pubis.

I: muscle fibers pass downward, backward& laterally & attached to **Pectineal line.**



A: Flexion, adduction, assists with medial rotation of the thigh at hip joint.

NS: Femoral nerve **+ obturator** Nerve **(occasionally).**


Rectus
Femoris
VL
VMO

5) Quadriceps femoris العضلة الرباعية الفخذية

- It forms the main bulk of the anterior thigh.

Consists of 4 heads, have a common tendon of insertion:

1-Rectus femoris(RF)

2-Vastus lateralis(VL)

3-Vastus medialis(VM)

4-Vastus intermedius(VI)

A-Rectus femoris (the only part crosses hip joint/bipennate)

O : 1)Stright head: from Ant. *Inferior*. Iliac. Spine.

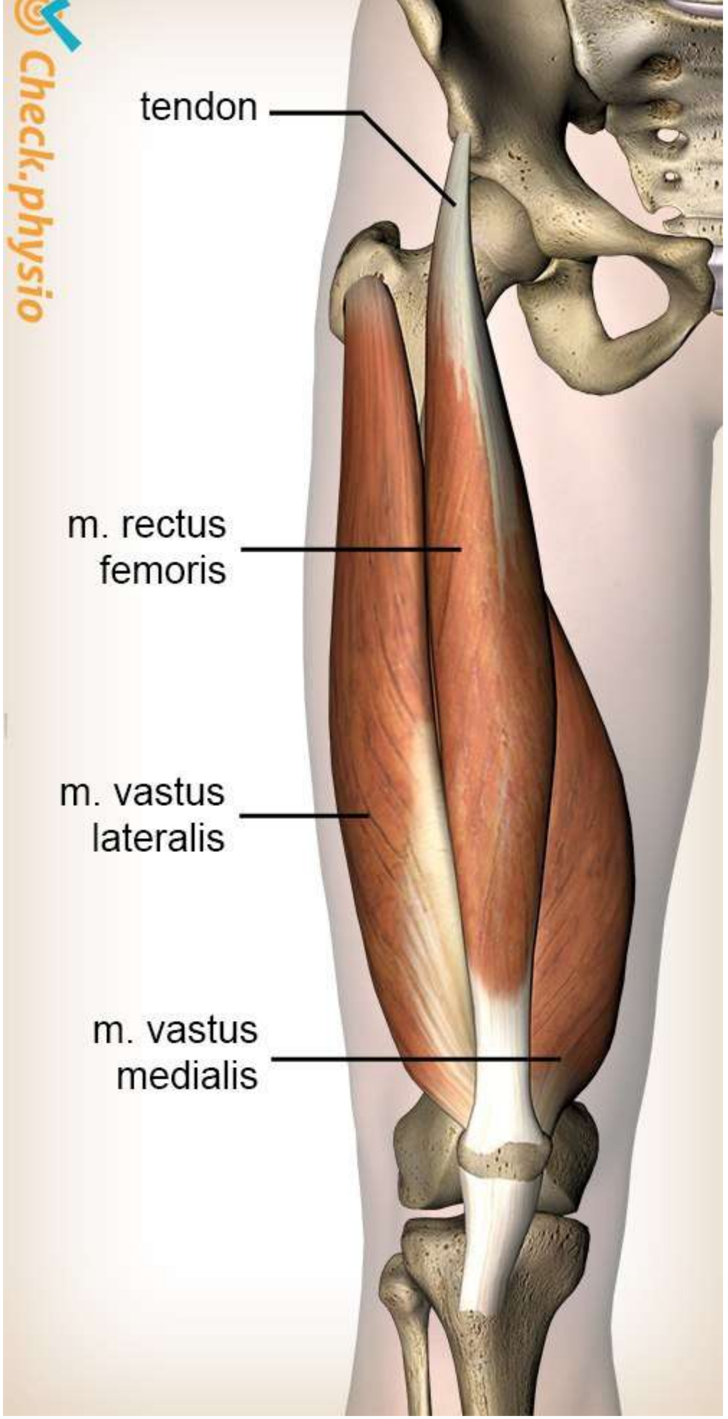
2)Reflected head: from ilium above acetabulum.

I: The two heads unite& form bipennate muscle, inserted into quadriceps tendon & so into patella.

A: **1-Flexes the thigh at hip joint.**

2-Extends the leg at the knee joint(**quadriceps mechanism**).

*****Acts on both joints.** **Other muscle?**



The attachment sites of the RECTUS FEMORIS muscles can vary between individuals.

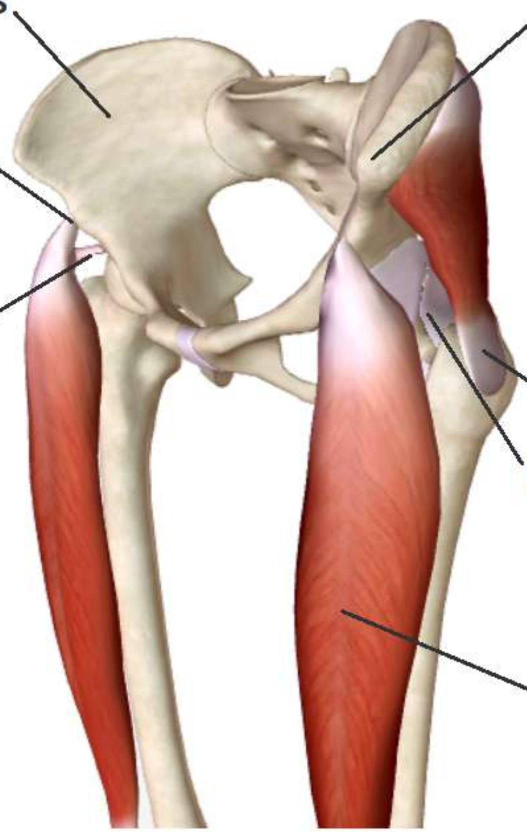


Image created using:
BioDigital
baselinehealing.com

Sartorius	Rectus femoris
Hip flexor	Hip flexor
O: A.sup.IS	O: A.Inferior IS (straight head)
N: femoral nerve.	Femoral nerve.
Acts on hip and knee joints.	Acts on hip and knee joints.
One head	2 heads
Flexor for both hip and knee joints.	**Flexor** for the hip/ **extensor** for the knee.

B-Vastus lateralis (largest component)

O: Intertrochanteric line/Greater trochanter / linea aspera of femur.

I: into quadriceps tendon & so into patella.

Ns: Femoral nerve.

C-Vastus Medialis

O: from intertrochanteric line & linea aspera of femur.

I: into quadriceps tendon & so into patella. Some fibers join capsule of knee joint ,strengthen it.

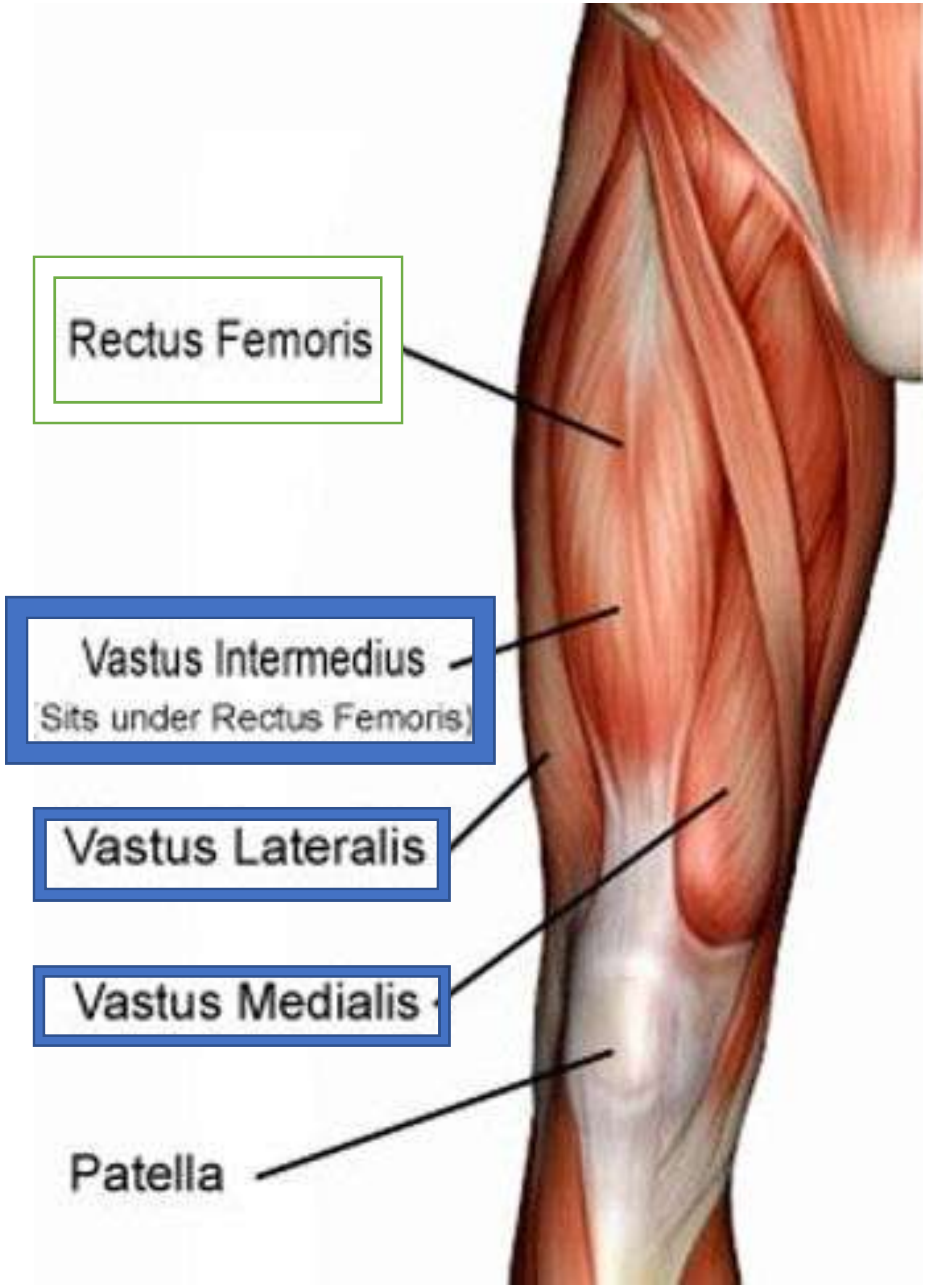
Ns: Femoral nerve.

D-Vastus intermedius (deep to R.Femoris)

O : from the anterior & lateral surfaces of the shaft of the femur.

I: fibers join the deep aspect of quadriceps tendon.

Ns: Femoral nerve.



❑ Insertion of quadriceps tendon:

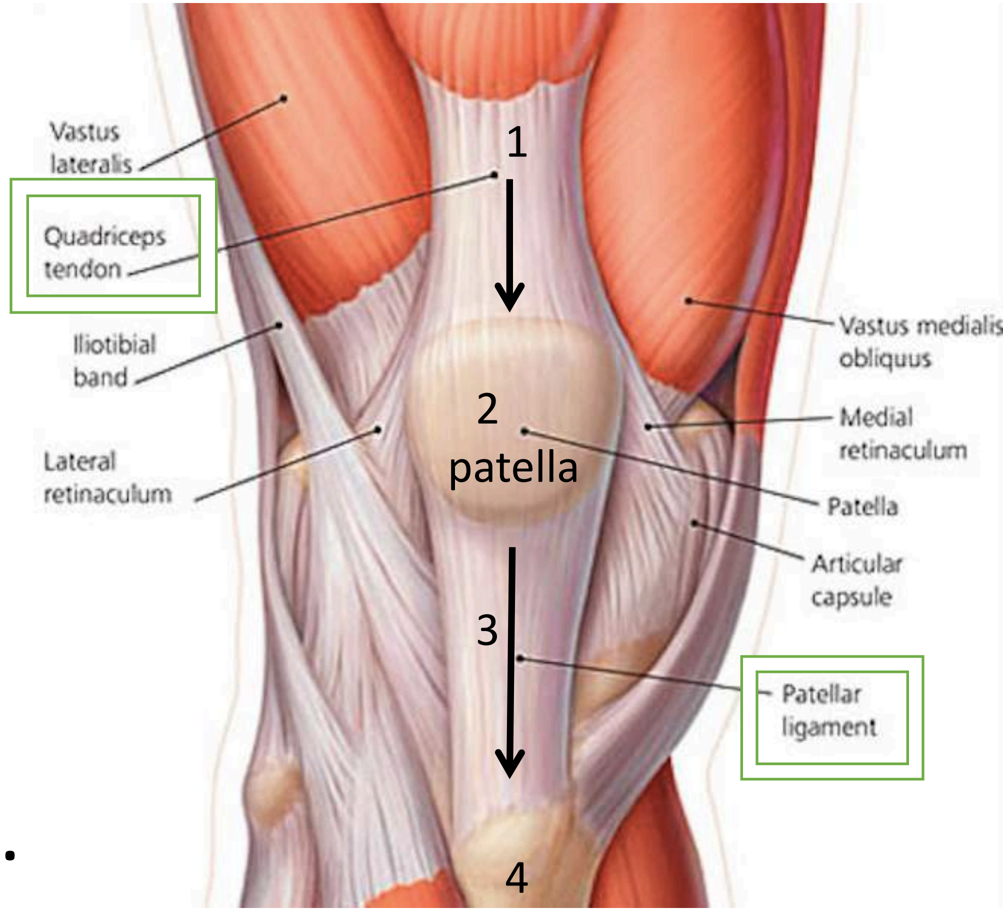
1- Via common tendon (quadriceps tendon)

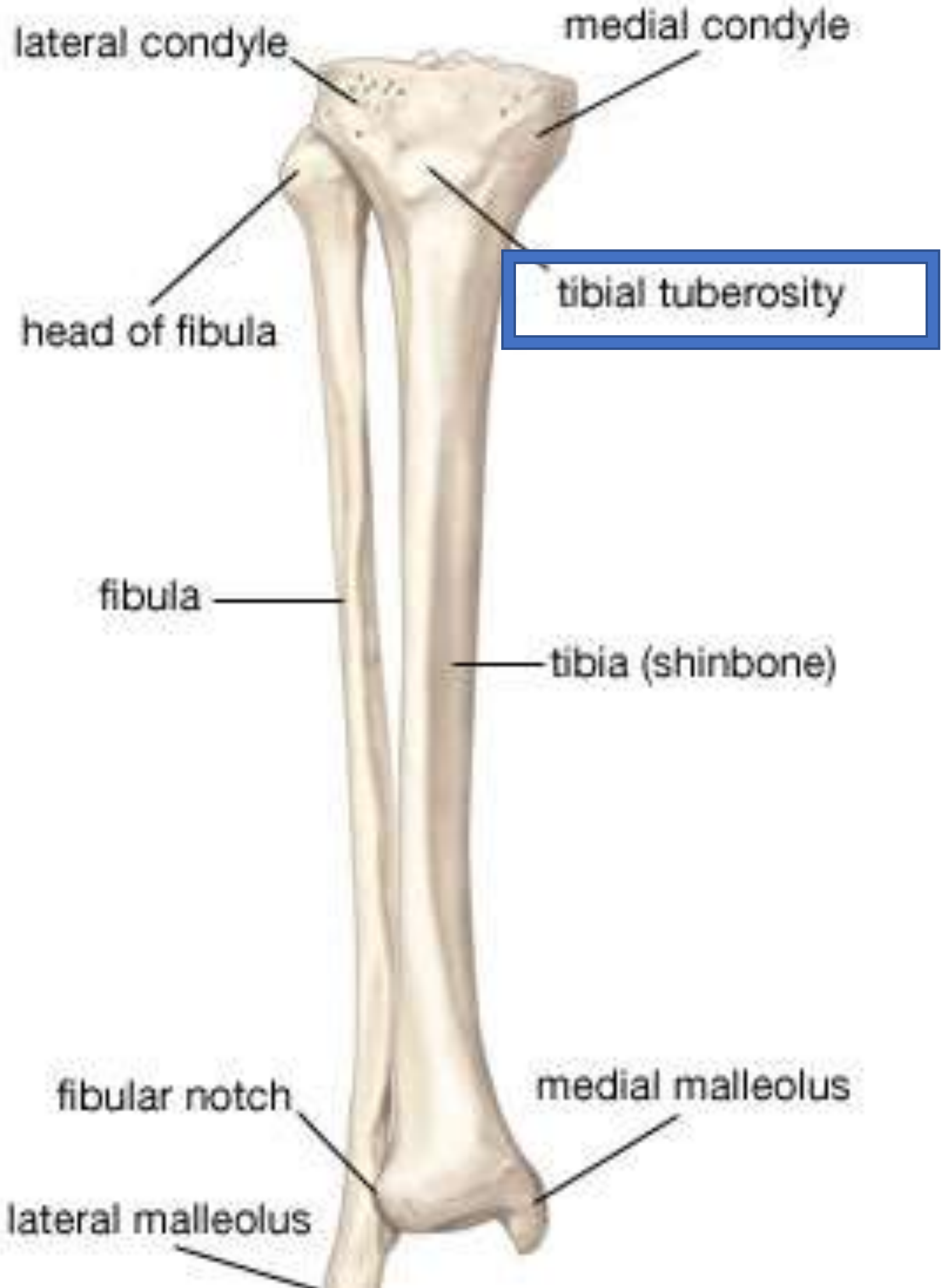
2- Attachment to base of patella (sesamoid bone)

3- indirectly via patellar ligament into **tibial tuberosity.**

Articularis genus of the knee? Inserted into synovial membrane of the knee joint. **Derivative of vastus intermedius.**

NB: medial& lateral vasti have extra attachment to tibia&patella via medial& lateral patellar retinacula.



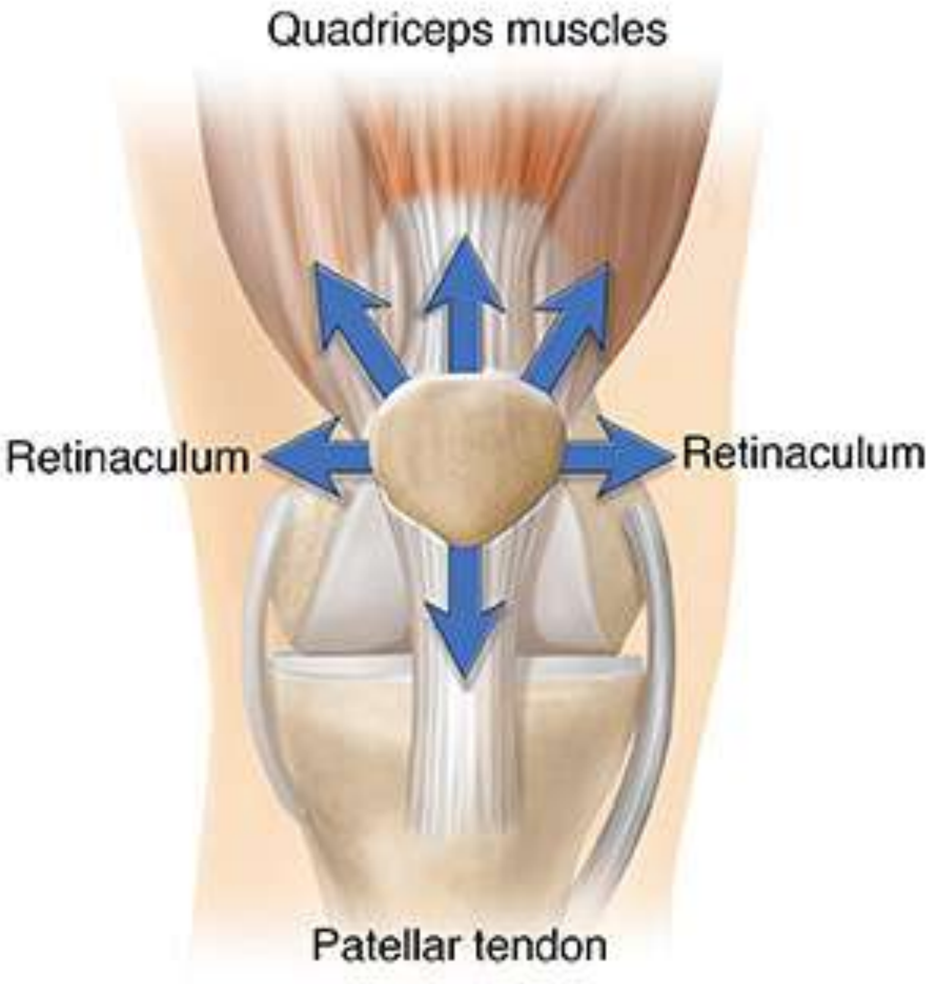

lateral condyle
medial condyle
head of fibula
tibial tuberosity
fibula
tibia (shinbone)
fibular notch
medial malleolus
lateral malleolus


Quadriceps mechanism
Hamstring muscles
©MMG 2008

Action of quadriceps femoris (quadriceps mechanism):

- ❖Because of their insertion, they are **powerful extensors of knee joint**.
- ❖They strengthen the knee joint; because some of tendinous fibers of Vast.lateralis and medialis form bands (retinacula) that join capsule of knee joint.
- ❖The lower fibers of **Vast.medialis** are almost horizontal, prevent patella from being pulled laterally during quadriceps contraction.
- ❖The **tone** of quadriceps muscle greatly strengthen the knee joint.
- ❖The rectus femoris also **flex the hip joint**.

Ligamentum Patellae: is a thick and strong ligament which carries insertion of the quadriceps femoris into the tibial tuberosity. Attached *superiorly* to apex of &post.surface of patella. *Inferiorly* attached to tibial tuberosity.




Femoral
groove
Patella


Hamstrings contract to bend the leg, quadriceps relax
Quadriceps contract to straighten the leg, hamstrings relax
Tendons join muscle to bone
Knee joint - where the femur and tibia meet

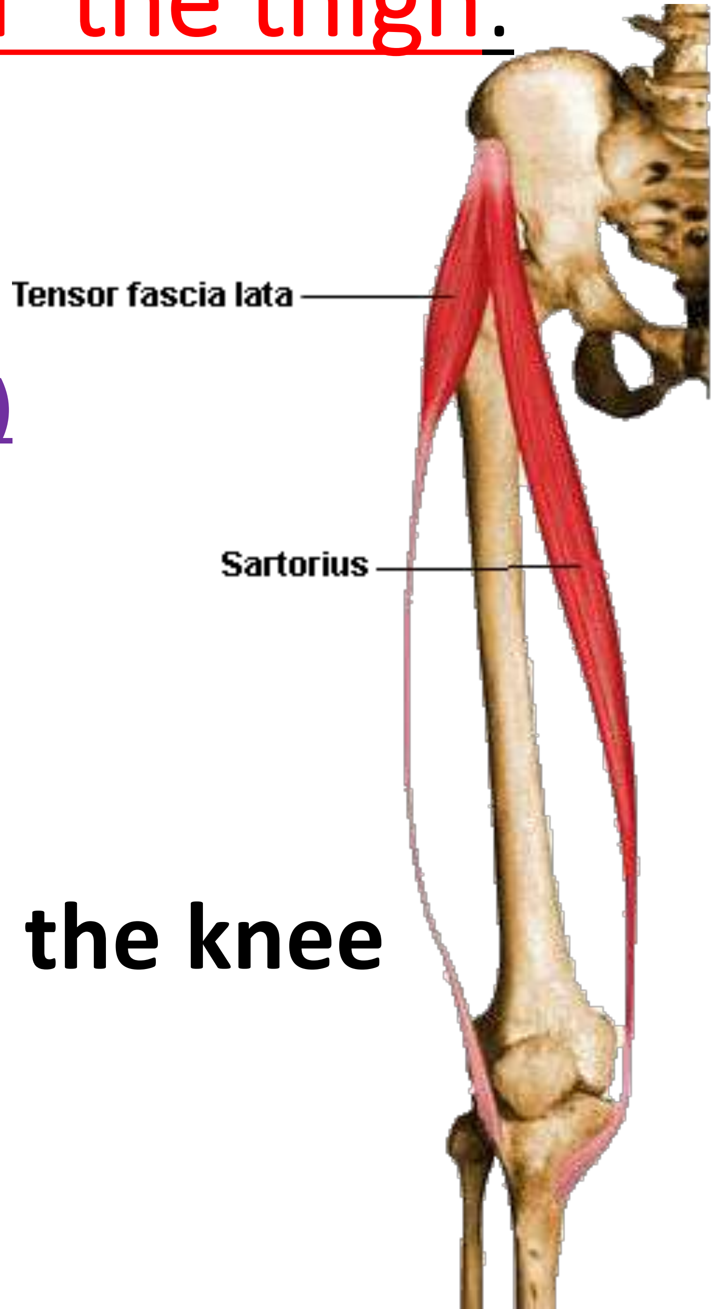
Two muscles of the anterior group act on **both hip and knee joint**

1-Sartorius; flexion of both.

2-Rectus femoris; flexion for hip

& extension for knee.

Muscles of the lateral Compartment of the thigh:



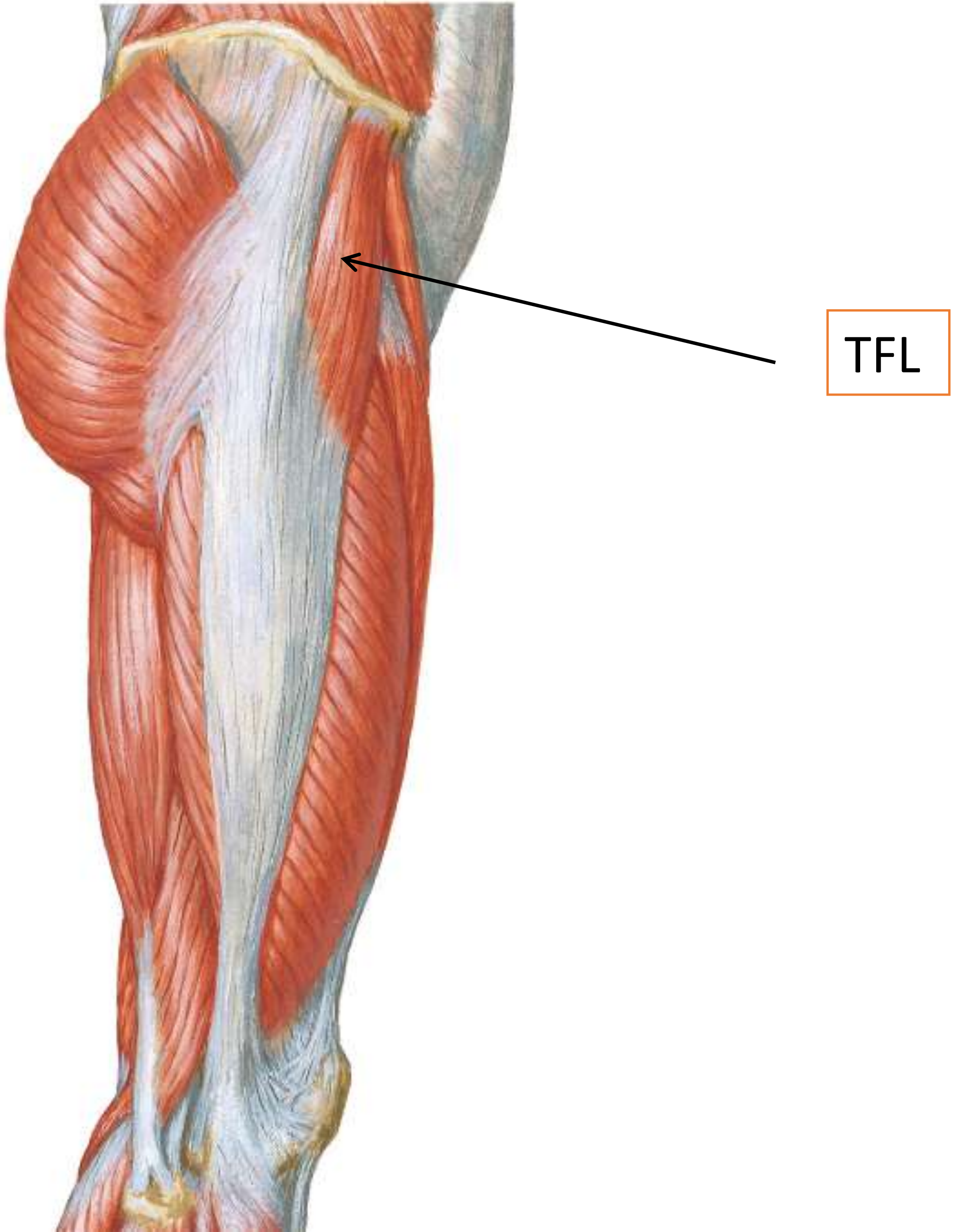
❖ Tensor fascia latae (fusiform M. lies on the lateral side of front of the thigh, enclosed in fascia lata)

O : from ASIS & iliac crest.

I : into Iliotibial tract.

N : *Superior gluteal nerve*.

A : assists gluteus maximus in extending the knee joint in standing position.


TFL

- All the muscles of anterolateral compartment of thigh are supplied by **femoral nerve except** psoas major and tensor fascia lata.

- Two muscles arise from **anterior superior iliac spine** of the hip bone:

  1- Sartorius 2- Tensor fascia lata.

- One muscle arise from **anterior inferior iliac spine** of the hip bone:

  Straight head of rectus femoris

Flexors of the thigh at the hip joint:

1) Pectineus.
2) Iliopsoas (iliacus + psoas).
3) Sartorius.
4) Rectus femoris.

(Formative assessment)

1-Which of the following muscles is a flexor for both hip and knee joints?

A) Rectus femoris

B) Sartorius

C) Pectineus

D) Vastus lateralis

2-Which of the following muscles is supplied by superior gluteal nerve?

A) Sartorius

B) Iliacus

C) Psoas major

D) Tensor fascia lata.

3- which of the following is inserted into lesser trochanter of the femur?

A) Iliacus

B) pectineus

C) Sartorius

D) Rectus femoris

4- which of the following muscles is flexor for the hip and extensor for the knee joint?

A) Sartorius

B) Rectus femoris

C) Vastus lateralis

D) Vastus medialis

At the hip joint, the sartorius performs all of the following actions except

A. Flexion
B. Lateral rotation
C. Medial rotation
D. Abduction

حال الطالبات في بداية دراسة الأطراف السفلية ☺

ليس خطوة واحدة عملاقة
التي حقَّقت **الإنجاز**
إنَّما مجموعة خطوات صغيرة